These are notes for Malek Abdesselam's Functional Analysis class for Fall 2026.

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Motivation

Consider a space V of (sufficiently nice) functions $f : \mathbb{R} \to \mathbb{R}$. Then, similar to how a vector in $\mathbb{R}^n$ can be denoted as

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix},$$

we may view the function $f : \mathbb{R} \to \mathbb{R}$ as a "column vector" with uncountable index. That is, $f = (f(x))_{x \in \mathbb{R}}$. If $L : V \to V$ is a linear map, then an analogue of the finite-dimensional matrix representation can be viewed as a kernel that f is integrated against; that is,

$$(Lf)(x) = \int_{\mathbb{R}} K(x,y) f(y)\, dy.$$

In the special case that $L = \mathrm{id}_V$, then the kernel $K(x,y)$ is given by the Dirac delta distribution, $K(x,y) = \delta(x-y)$. The space of these distributions is the primary object of study in Fourier analysis.

Notation

- The natural numbers are given by $\mathbb{N} = \{0, 1, 2, 3, \ldots\}$.
- The strictly positive natural numbers are given by $\mathbb{Z}_{>0} = \{1, 2, 3, \ldots\}$.

- A multi-index is a tuple $\alpha \in \mathbb{N}^n$. We let
$$|\alpha| = \alpha_1 + \cdots + \alpha_n.$$
If $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$, then we will write
$$x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$$
and
$$\alpha! = \alpha_1! \cdots \alpha_n!$$
for the corresponding values.
If $f : \mathbb{R}^n \to \mathbb{R}$ is any function, then we will write
$$\partial^\alpha f = \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \cdots \partial x_n^{\alpha_n}}.$$
Operations and comparators on multi-indices will be taken componentwise.
- If A is a set, we will write $\mathcal{P}(A)$ for the power set of A. If A and B are sets, then the set $B^A \subseteq \mathcal{P}(A \times B)$ is the set of all functions $f : A \to B$.
We will write $\mathcal{P}_{\text{fin}}(X)$ to consist of all finite subsets subsets of X.
- If $x \in \mathbb{R}^n$ is given by $(x_1, \ldots, x_n)$, we will define the bracket
$$\begin{aligned}\langle x \rangle &:= \sqrt{1 + x_1^2 + \cdots + x_n^2} \\ &= \sqrt{1 + |x|^2}.\end{aligned}$$
In the particular case of $n = 1$, we will write $\langle k \rangle = \sqrt{1 + k^2}$ for any $k \in \mathbb{N}$. Note here that $|\cdot|$ denotes the norm on Euclidean space.
- If $n \in \mathbb{N}$, then
$$[n] = \{1, 2, \ldots, n\},$$
where we have $[0] = \emptyset$.
- If A is a finite set, then $|A|$ is the cardinality of A.
- Generally we will denote by $\mathbb{K}$ for either $\mathbb{R}$ or $\mathbb{C}$.
- If $V_1, \ldots, V_n, W$ are topological vector spaces, we will write $\mathcal{L}_n(V_1, \ldots, V_n; W)$ to denote the space of (jointly) continuous multilinear maps $V_1 \times \cdots \times V_n \to W$. If $n = 1$, we will just write $\mathcal{L}(V, W)$.
- The Borel σ-algebra on a topological space $(X, \mathcal{T}_X)$ is given by $\mathcal{B}_X$.
- If X is a compact metric space, the space $C(X)$ is the Banach space of continuous functions with values in the assumed field.

Locally Convex Topological Vector Spaces

Recall that if X is a set, then $\mathcal{T} \subseteq \mathcal{P}(\mathcal{P}(X))$ is a topology on X if it satisfies:

- $\emptyset, X \in \mathcal{T}$;
- $U_1, U_2 \in \mathcal{T}$ implies $U_1 \cap U_2 \in \mathcal{T}$;
- $\{U_i : i \in I\} \subseteq \mathcal{T}$ implies $\bigcup_{i \in I} U_i \in \mathcal{T}$.

Definition: Definitions

We say $(V, \mathcal{T})$ is a *topological vector space* (over $\mathbb{K}$) if V is a $\mathbb{K}$-vector space whose topology is such that the maps

$$\begin{aligned} V \times V &\to V \\ (u, v) &\mapsto u + v \end{aligned}$$

and

$$\begin{aligned} \mathbb{K} \times V &\to V \\ (\lambda, v) &\mapsto \lambda v \end{aligned}$$

are continuous with respect to the product topology. Note that $\mathbb{K}$ is equipped with its traditional topology.

The morphisms of topological vector spaces are the *continuous* linear maps. We say $V \cong W$ are isomorphic as topological vector spaces if there is $\varphi\colon V \to W$ that is bijective, linear, and continuous, and φ^{-1} is also continuous.

Definition: Let V be a $\mathbb{K}$-vector space. We say $\rho\colon V \to [0, \infty)$ is a *seminorm* if

$$\begin{aligned} \rho(v + w) &\leq \rho(v) + \rho(w) \\ \rho(\lambda v) &= |\lambda| \rho(v) \end{aligned}$$

for any $v, w \in V$ and $\lambda \in \mathbb{K}$.

Remark: The only difference between a seminorm and a norm is that $\rho(v) = 0$ need not imply that $v = 0$. A norm is a seminorm.

We will let $\mathrm{SN}(V)$ be the set of all seminorms of V.

Definition: Let V be a $\mathbb{K}$-vector space. Let $\mathcal{N} \subseteq \mathrm{SN}(V)$. Define $\mathcal{T}(\mathcal{N})$ to be a topology with basic open sets $U(x, F, \varepsilon) \subseteq V$ given by

$$\begin{aligned} U(x, F, \varepsilon) &= \{y \in V : \rho(y - x) < \varepsilon \text{ for all } \rho \in F\} \\ &= \bigcap_{\rho \in F} U(x, \rho, \varepsilon). \end{aligned}$$

for any $x \in V$, $F \subseteq \mathcal{N}$ finite, and $\varepsilon > 0$.

Remark: I prefer to use the subbasic open sets

$$U(x, \rho, \varepsilon) = \{y \in V : \rho(y - x) < \varepsilon\}$$

and write intersections.

We observe that the subsets of the form $U(x, \{\rho\}, \varepsilon)$ as ρ ranges across $\mathcal{N}$ forms a subbase for the topology of V.

Proposition: If $F \subseteq \mathcal{N}$ is finite, $\varepsilon > 0$, and $x \in V$, then

$$U(x, F, \varepsilon) \in \mathcal{T}(\mathcal{N}).$$

Proof. Suppose $y \in U(x, F, \varepsilon)$. Our goal is to find $\eta > 0$ such that $U(y, F, \eta) \subseteq U(x, F, \varepsilon)$. Suppose $z \in U(y, F, \eta)$. If $\rho \in F$, then

$$\begin{aligned} \rho(z - x) &\leq \rho(z - y) + \rho(y - x) \\ &< \eta + \rho(y - x). \end{aligned}$$

Therefore, we will define η to be small enough such that $\eta + \max_{\rho \in F} \rho(y - x) < \varepsilon$. This gives the desired result. □

Definition: A topological space is called a locally convex topological vector space (locally convex topological vector space) if the topology is generated by a family of seminorms $\mathcal{N}$ with the property that

$$\bigcap_{\rho\in\mathcal{N}}\{v : \rho(v) = 0\} = \{0\}.$$

A family $\mathcal{N}$ of seminorms that satisfies this property is said to be *separating*.

Example: If H is a Hilbert space with orthonormal basis $\{e_n : n \in \mathbb{N}\}$, then the seminorms $v \mapsto |\langle v, e_i\rangle|$ are separating.

Theorem: If V is a $\mathbb{K}$-locally convex topological vector space and $\mathcal{N}$ is a collection of seminorms that defines the topology on V, then the pair $(V, \mathcal{T}(\mathcal{N}))$ is Hausdorff if and only if $\mathcal{N}$ separates the points of V.

Proof. Assume $\mathcal{N}$ separates the points of V. Suppose $v \neq w$ are distinct elements of V. Then, since $v - w \neq 0$, there is some $\rho \in \mathcal{N}$ such that $\rho(v-w) = \varepsilon_0 > 0$. Consider the finite set $F = \{\rho\}$, and choose some $0 < \varepsilon < \varepsilon_0/2$. Then, we claim that the basic open sets $U(v, F, \varepsilon)$ and $U(w, F, \varepsilon)$ are disjoint. This follows from the fact that if $x \in U(v, F, \varepsilon)$, then $\varepsilon_0 = \rho(v-w) \leq \rho(v-x) + \rho(x-w)$, so that $\rho(w-x) \geq \varepsilon_0 - \varepsilon > \varepsilon_0 - \varepsilon_0/2 > \varepsilon$, yielding the desired result.

Now, suppose $\mathcal{N}$ does not separate the points of V. Then, there are some distinct $v \neq w$ such that for all $\rho \in \mathcal{N}$, we have $\rho(v-w) = 0$. In particular, observe that for all finite subsets $F \subseteq \mathcal{N}$ and all $\varepsilon > 0$, we have that w is contained in $U(v, F, \varepsilon)$, meaning that there are no basic open sets that separate v and w. Since any open subset of V contains a basic open subset, it follows that for any open subsets $v \in U$ and $w \in U'$, we have some finite F such that $U(v, F, \varepsilon) \subseteq U$, but this also has $w \in U(v, F, \varepsilon)$, implying that $U \cap U' \neq \emptyset$. Thus, V is not Hausdorff. □

Examples

Remark: If we remove the assumption that the family $\mathcal{N}$ is separating, then the topology induced by the trivial seminorm $\rho\colon V \to [0, \infty)$ is the indiscrete topology on V. Similarly, there is a family of seminorms that induces the discrete topology on V.

(i) Let $(V, \|\cdot\|)$ be a normed vector space over $\mathbb{K}$. Examples of these include Banach spaces such as $L^p(X, \Sigma, \mu)$, where Σ is a σ-algebra and μ is a positive measure on the measurable space (X, Σ), and $p \in [1, \infty]$.

There is a metric on this normed space $d(x, y) = \|x - y\|$, which gives rise to the metric topology $\mathcal{T}(d)$. We have $U \in \mathcal{T}(d)$ if and only if $U \subseteq V$ is such that for any $x \in U$, there exists $\varepsilon > 0$ such that

$$\begin{aligned}\{y \in V : d(x, y) < \varepsilon\} &=: U(x, \varepsilon)\\ &\subseteq U.\end{aligned}$$

This is a locally convex space with seminorm $\mathcal{N} = \{\|\cdot\|\}$. There are two possibilities for finite subsets $F \subseteq \{\|\cdot\|\}$, meaning that if $F = \emptyset$, then $U(x, F, \varepsilon) = U(x, \emptyset, \varepsilon) = V$, while if $F = \{\|\cdot\|\}$, then

$$\begin{aligned}U(x, F, \varepsilon) &= \{y \in V : \rho(x-y) < \varepsilon \text{ for all } \rho \in F\}\\ &= \{y \in V : \|x - y\| < \varepsilon\}\\ &= U(x, \varepsilon).\end{aligned}$$

(ii) Define $s(\mathbb{N}, \mathbb{K})$ to be the space of sequences $x = (x_n)_{n\geq 0}$ in $\mathbb{K}^{\mathbb{N}}$ such that for all $k \geq 0$, $\sup_{n\geq 0}\langle n\rangle^k |x_n| < \infty$, where $\langle n\rangle = \sqrt{1+n^2}$ is the bracket. This is the space of *rapidly decaying sequences* in $\mathbb{K}$. In essence, elements of $s(\mathbb{N}, \mathbb{K})$ decay faster than any inverse power of the index n.

We define the seminorms

$$\rho_k(x) = \sup_{n\geq 0}\langle n\rangle^k |x_n|.$$

In fact, the $\{\rho_k : k \in \mathbb{N}\}$ are norms. Let

$$\mathcal{N} := \{\rho_k : k \in \mathbb{N}\}$$

We obtain a locally convex on $s(\mathbb{N}, \mathbb{K})$ with topology induced by the family $\mathcal{N}$.

Later, we will see that $(s(\mathbb{N}, \mathbb{K}), \mathcal{T}(\mathcal{N}))$ is *not* a normed space. That is, there does not exist a norm $\|\cdot\|$ such that $\mathcal{T}(\mathcal{N}) = \mathcal{T}(\{\|\cdot\|\})$.

(iii) Take $V = C([0,\infty), \mathbb{R})$ to be the set of all real-valued continuous functions on $[0,\infty)$ with pointwise addition and scalar multiplication. This is a real vector space.

At first, we may think $\|f\| = \sup_{t\in\mathbb{R}}|f(t)|$ would define a norm, but this is not satisfactory since the functions in V are not necessarily bounded.

Therefore, we want to restrict to compact subsets. Take

$$\rho_k(f) := \sup_{0\le t\le k} |f(t)|.$$

Notice that the ρ_k are not norms (use Urysohn's Lemma or your favorite piecewise function to construct one). However, the family $\mathcal{N} = \{\rho_k : k \ge 0\}$ is a separating family.

Consider the set of functions $V_0 = \{f \in V : f(0) = 0\}$. Then, $V_0 \le V$ is a closed subset.

> *Proof.* Consider the linear functional $\mathrm{ev}_0 \colon V \to \mathbb{R}$ given by $f \mapsto f(0)$. We claim that this linear functional is continuous.
>
> Let $U \subseteq \mathbb{R}$ be open. Let $z \in U$, and let $\varepsilon > 0$ be such that $U(z,\varepsilon) := \left\{y \in \mathbb{R} : \left|z - y\right| < \varepsilon\right\}$ is contained in U. Let $f \in \mathrm{ev}_0^{-1}(U(z,\varepsilon))$, and set $\varepsilon_0 = |z - f(0)|$. To show that $\mathrm{ev}_0^{-1}(U(z,\varepsilon))$ is open, it suffices to find some seminorm ρ_k and some $\delta > 0$ such that $U(f, \rho_k, \delta) \subseteq \mathrm{ev}_0^{-1}(U(z,\varepsilon))$.
>
> In fact, we may choose $k = 1$ and $\delta = \varepsilon - \varepsilon_0$. Suppose $g \in U(f, \rho_1, \delta)$. Then, we have that
>
> $$\begin{aligned} |g(0) - z| &\le |g(0) - f(0)| + |f(0) - z| \\ &\le \sup_{t\in[0,1]} |g(t) - f(t)| + \varepsilon_0 \\ &< \delta + \varepsilon_0 \\ &= \varepsilon. \end{aligned}$$
>
> In particular, Thus, $U(f, \rho_1, \delta) \subseteq \mathrm{ev}_0^{-1}(U(z,\varepsilon))$, so that $\mathrm{ev}_0^{-1}(U(z,\varepsilon))$ is open as any element in this set admits a basic open neighborhood contained in the set. Therefore, ev_0 is continuous.
>
> Since $V_0 = \mathrm{ev}_0^{-1}(\{0\})$, it follows that V_0 is closed since $\{0\} \subseteq \mathbb{R}$ is closed. □

We claim that the subspace topology on V_0 is given by $\mathcal{T}_0 = \mathcal{T}\left(\left\{\rho_k|_{V_0} : k \ge 0\right\}\right)$.

> *Proof.* First, we will show that, given a subbasic open neighborhood $U = V_0 \cap U(x, \rho_k, \varepsilon)$ in the subspace topology of V_0 and any element y in this subbasic open neighborhood, there is an open neighborhood $V \in \mathcal{T}\left(\left\{\rho_k|_{V_0} : k \ge 0\right\}\right)$ such that such that $y \in V \subseteq U$.
>
> If $y \in U$, we observe that $y \in U(x, \rho_k, \varepsilon)$. In particular, since these form a subbase for $\mathcal{T}(\{\rho_k : k \ge 0\})$, it follows that there is some finite collection $F = \left\{\rho_{i_j} : j = 1, \dots, m\right\}$ of seminorms and some $\delta > 0$ such that
>
> $$U(y, F, \delta) = \bigcap_{j=1}^{m} U\left(y, \rho_{i_j}, \delta\right).$$

Since $U\left(y, \rho_{i_j}|_{V_0}, \delta\right) \subseteq U\left(y, \rho_{i_j}, \delta\right)$ for each j, it follows that we have

$$\bigcap_{j=1}^{m} U\left(y, \rho_{i_j}|_{V_0}, \delta\right) \subseteq V_0 \cap \left(\bigcap_{j=1}^{m} U\left(y, \rho_{i_j}, \delta\right)\right),$$

whence setting V to equal the left hand side yields the desired result. Therefore, $\mathcal{T}\left(\left\{\rho_k|_{V_0} : k \geq 0\right\}\right)$ refines the subspace topology $\mathcal{T}_0$.

In the reverse direction, observe that if $y \in U\left(x, \rho_j|_{V_0}, \varepsilon\right)$ in V_0, then $y \in U\left(x, \rho_j, \varepsilon\right)$, so there is some $\delta > 0$ and finite $F = \left\{\rho_{i_j} : j = 1, \ldots, m\right\}$ such that

$$\begin{aligned} U(y, F, \varepsilon) &= \bigcap_{j=1}^{m} U\left(y, \rho_{i_j}, \delta\right) \\ &\subseteq U\left(x, \rho_j, \varepsilon\right), \end{aligned}$$

but we notice then that $U\left(x, \rho_j, \varepsilon\right) \cap V_0 \subseteq U\left(x, \rho_j|_{V_0}, \varepsilon\right)$, so that $V_0 \cap U(y, F, \varepsilon) \subseteq U\left(x, \rho_j|_{V_0}, \varepsilon\right)$, so that $\mathcal{T}_0$ refines $\mathcal{T}\left(\left\{\rho_k|_{V_0} : k \geq 0\right\}\right)$. Thus, these topologies are equal. □

We may consider $\mathcal{B}(\mathcal{T}_0)$ to be the Borel σ-algebra on V_0. Brownian motions are a special class of probability measure on $(V_0, \mathcal{B}(\mathcal{T}_0))$. Specifically, they are the probability measures such that for every $n \geq 0$, partition $0 < t_1 < t_2 < \cdots < t_n$, the random variables

$$\begin{aligned} X_1 &= f \mapsto f(t_1) \\ X_2 &= f \mapsto (f(t_2) - f(t_1)) \\ X_3 &= f \mapsto (f(t_3) - f(t_2)) \\ &\vdots \\ X_n &= f \mapsto (f(t_n) - f(t_{n-1})) \end{aligned}$$

are independent centered Gaussian random variables with variances $t_1, t_2 - t_2, \ldots, t_n - t_{n-1}$ for an arbitrary f.

Similar to the case of $s(\mathbb{N}, \mathbb{K})$, the space $(V_0, \mathcal{T}_0)$ is not a normed space.

(iv) Consider the vector space $V = \mathbb{R}^{\mathbb{N}}$, which is the space of all real-valued sequences. Let $\mathcal{T}$ be the product topology on $\mathbb{R}^{\mathbb{N}}$. We are curious as to whether $(V, \mathcal{T})$ is a locally convex topological vector space.

Consider the function $d\colon V \times V \to [0, \infty)$ defined by

$$d(x, y) = \sum_{n=0}^{\infty} 2^{-n} \min\left(1, \left|x_n - y_n\right|\right).$$

Then, we have that d is a distance metric, and $\mathcal{T} = \mathcal{T}(d)$.

Proof. We start by showing that if $x \in V$ is any element and $\varepsilon > 0$, then $U(x, \varepsilon)$ is open in $\mathcal{T}$. Translating, we take $x = 0$ (since the metric d is translation-invariant), so that we are now seeking to show that $U(0, \varepsilon)$ is open. This is equivalent to showing that if $x \in U(0, \varepsilon)$, then there is a finite collection of subbasic open sets $\pi_j^{-1}\left(U_j\right)$ whose intersection W contains x and is contained in $U(0, \varepsilon)$.

To start, we observe that there is some index n_0 such that

$$\sum_{n=n_0+1}^{\infty} 2^{-n} < \varepsilon/2.$$

We then select $U_j = \left(x_j - \varepsilon/4, x_j + \varepsilon/4\right)$ for each $j = 0, \dots, n_0$. It then follows that $x \in W$, and we have for any $y \in W$ that

$$\begin{aligned} d(x,y) &= \sum_{j=0}^{\infty} 2^{-j} \min\left(1, \left|x_j - y_j\right|\right) \\ &\leq \sum_{j=0}^{n_0} 2^{-j} \min\left(1, \left|x_j - y_j\right|\right) + \sum_{j=n_0+1}^{\infty} 2^{-j} \\ &< \sum_{j=0}^{n_0} 2^{-j}(\varepsilon/4) + \varepsilon/2 \\ &\leq \varepsilon, \end{aligned}$$

whence $x \in W \subseteq U(0, \varepsilon)$.

Now, consider a basic open set $W = \bigcap_{j=0}^{n_0} \pi_j^{-1}\left(U_j\right)$. Let $x \in W$. Let $\delta_0, \dots, \delta_{n_0}$ be such that $\left(x_j - \delta_j, x_j + \delta_j\right) \subseteq U_j$ for each $j = 1, \dots, n$, and set $\delta = 2^{-n_1} \min\left(\delta_0, \dots, \delta_{n_0}\right)$, where n_1 is any integer strictly larger than n_0.

Notice then that if $y \in U(x, \delta)$, then we have for each $j = 0, \dots, n_0$ that

$$\begin{aligned} \left|x_j - y_j\right| &< 2^j \delta \\ &\leq 2^{n_0} 2^{-n_1} \min\left(\delta_0, \dots, \delta_{n_0}\right) \\ &< \min\left(\delta_0, \dots, \delta_{n_0}\right) \\ &\leq \delta_j. \end{aligned}$$

In particular, this means we must have $U(x, \delta) \subseteq W$.

Therefore, we have that $\mathcal{T}$ and $\mathcal{T}(d)$ generate the same topology on V. □

In particular, we may consider the seminorms

$$\rho_n(x) = |x_n|,$$

and $\mathcal{N} = \{\rho_n : n \geq 0\}$. Then, we claim that $\mathcal{T} = \mathcal{T}(\mathcal{N})$.

Proof. A basic open neighborhood of $x \in V$ in $\mathcal{T}(\mathcal{N})$ is given by the sets

$$U(x, F, \varepsilon) = \bigcap_{j \in F} \left\{(y_n)_n : \left|x_j - y_j\right| < \varepsilon\right\}.$$

Meanwhile, a basic open neighborhood of x in $\mathcal{T}$ is given by the sets

$$U'(x, F, \varepsilon) = \bigcap_{j \in F} \pi_j^{-1}\left(\left(x_j - \varepsilon, x_j + \varepsilon\right)\right).$$

In fact, these are the same subsets of V by the definition of the projection map $\pi_j \colon V \to \mathbb{R}$ taking $(x_n)_n \mapsto x_j$.

Therefore, if we have $y \in U(x, F, \varepsilon)$ in $\mathcal{T}(\mathcal{N})$, then if we define

$$\delta = \frac{1}{2} \min\left(\left\{\left|\varepsilon - y_j\right| : y_j \in F\right\}\right),$$

we have that $U'(y, F, \delta) \subseteq U(x, F, \varepsilon)$. Similarly, if $y \in U'(x, F, \varepsilon)$, then setting δ to equal the same as above, we have $U(y, F, \varepsilon) \subseteq U'(x, F, \varepsilon)$.

This gives $\mathcal{T} = \mathcal{T}(\mathcal{N})$. □

In particular, V is not a normed space.

(v) Consider $c_{00}(\mathbb{N},\mathbb{K}) \cong \mathbb{K}[x]$ to consist of all the finitely supported sequences $(x_n)_{n\geq 0}$. This is a vector space. Consider the seminorms $\mathcal{N} = \mathrm{SN}(V)$, and the locally convex space $(V,\mathcal{T}(\mathcal{N}))$. Then, this is a locally convex space that is not metrizable. The topology $\mathcal{T}(\mathcal{N})$ is known as the finest locally convex topology.

General Properties of Topological Vector Spaces

Definition: Let $(V,\mathcal{T})$ be a $\mathbb{K}$-TVS, and let $\tau \in \mathrm{SN}(V)$. Then, τ is a *continuous seminorm* for $(V,\mathcal{T})$ if $\tau\colon V \to [0,\infty) \subseteq \mathbb{R}$ is continuous.

Proposition: Let V be a $\mathbb{K}$-vector space, $\mathcal{N} \subseteq \mathrm{SN}(V)$ define a locally convex topology on V. If $\tau \in \mathrm{SN}(V)$, then τ is continuous for $(V,\mathcal{T}(\mathcal{N}))$ if and only if there exists $C \geq 0$, $n \in \mathbb{N}$, and $\rho_1,\dots,\rho_n \in \mathcal{N}$ such that

$$\tau(v) \leq C \max\Big\{\rho_j(v) : j = 1,\dots,n\Big\} \tag{$*$}$$

or

$$\tau(v) \leq C \sum_{j=1}^{n} \rho_j(v).$$

Proof. Suppose τ is continuous. Then, $[0,1)$ is open in $[0,\infty)$, so $\tau^{-1}([0,1))$ is open in $\mathcal{T}(\mathcal{N})$ with $0 \in \tau^{-1}([0,1))$.

So, by definition, there exists $\varepsilon > 0$ and $\rho_1,\dots,\rho_n$ such that

$$\bigcap_{j=1}^{n} U\Big(0,\rho_j,\varepsilon\Big) \subseteq \tau^{-1}([0,\infty)).$$

That is, for any $i \in [n]$, we have that if $\rho_i(x) < \varepsilon$, then $\tau(x) < 1$.

Let $x \in V$ and $\lambda > 0$. Then,

$$\begin{aligned}\tau(x) &= \tau\Big(\lambda\lambda^{-1}x\Big)\\ &= \lambda\tau\Big(\lambda^{-1}x\Big)\\ &< \lambda\end{aligned}$$

if for every i, we have $\rho_i\Big(\lambda^{-1}x\Big) < \varepsilon$.

Define $J = \Big\{\lambda > 0 : \rho_i\Big(\lambda^{-1}x\Big) < \varepsilon \text{ for all } i \in [n]\Big\}$. Rewriting, we have

$$\begin{aligned}J &= \left\{\lambda > 0 : \frac{1}{\varepsilon}\rho_i(x) < \lambda \text{ for all } i \in [n]\right\}\\ &= \bigcap_{i=1}^{n}\left(\frac{1}{\varepsilon}\rho_i(x),\infty\right),\end{aligned}$$

giving $\inf(J) = \frac{1}{\varepsilon}\max\{\rho_i(x) : 1 \leq i \leq n\}$. Therefore, if $\lambda \in J$, then $\tau(x) \leq \lambda$, whence $\tau(x) \leq \inf(J)$. Therefore, we have shown that $\tau(x) \leq \frac{1}{\varepsilon}\max\{\rho_i(x) : 1 \leq i \leq n\}$.

In the reverse direction, suppose $(*)$ holds. Let $x \in V$, $\rho_1,\dots,\rho_n \in \mathcal{N}$, and let $\eta > 0$. Let

$$y \in \bigcap_{i=1}^{n} U(x,\rho_i,\eta).$$

By the reverse triangle inequality, we have

$$\Big|\tau(y) - \tau(x)\Big| \leq \tau(y-x)$$

$$< \varepsilon,$$

but

$$\begin{aligned}\tau(y-x) &\leq C\max\{\rho_i(y-x) : 1 \leq i \leq n\} \\ &\leq C\eta \\ &< \varepsilon\end{aligned}$$

if η is small enough. Thus, we have for all $y \in \bigcap_{j=1}^{n} U\Big(x, \rho_j, \eta\Big)$, then $\Big|\tau(y) - \tau(x)\Big| < \varepsilon$, whence τ is continuous at x. □

Remark: If $\tau \in \mathcal{N}$, then τ is continuous with respect to $\mathcal{T}(\mathcal{N})$.

Proposition: Let $(V, \mathcal{T}(\mathcal{N}))$ be a locally convex topological vector space. Then, $U \subseteq V$ is open if and only if for all $x \in U$, there is some continuous seminorm τ and some $\varepsilon > 0$ such that $U(x, \tau, \varepsilon) \subseteq U$.

Theorem (Continuity of Multilinear Maps on locally convex topological vector spaces)**:** Suppose $n \geq 1$, and $V_1, \ldots, V_n, W$ are $\mathbb{K}$-locally convex topological vector spaces, and let $M\colon\ V_1 \times \cdots \times V_n \to W$ be a $\mathbb{K}$-multilinear map.

Then, M is (jointly) continuous if and only if for all continuous seminorm τ on W, there exist $\rho_1, \ldots, \rho_n$ continuous seminorms on $V_1, \ldots, V_n$ respectively such that for all tuples $(x_1, \ldots, x_n) \in V_1 \times \cdots \times V_n$,

$$\tau(M(x_1, \ldots, x_n)) \leq \rho(x_1) \cdots \rho(x_n). \tag{†}$$

Proof. Suppose M is (jointly) continuous. That is, M is equipped with the product topology. Since τ is a continuous seminorm, we have that τ is a continuous map from $W \to [0, \infty)$, we have $\tau^{-1}([0,1))$ is open in W. Moreover, $M(0, \ldots, 0) = 0 \in \tau^{-1}([0,1))$. Since M is continuous at $(0, \ldots, 0)$, there is a neighborhood $U_1 \times \cdots \times U_n$ in $V_1 \times \cdots \times V_n$ where each U_i is a neighborhood of 0 in V_i, such that $M(U_1 \times \cdots \times U_n) \subseteq \tau^{-1}([0,1))$.

Let $(x_1, \ldots, x_n) \in V_1 \times \cdots \times V_n$, and let $\lambda_1, \ldots, \lambda_n > 0$. We have

$$\begin{aligned}\tau(M(x_1, \ldots, x_n)) &= \tau\Big(M\Big(\lambda_1\lambda_1^{-1}x_1, \ldots, \lambda_n\lambda_n^{-1}x_n\Big)\Big) \\ &= \lambda_1 \cdots \lambda_n \tau\Big(M\Big(\lambda_1^{-1}x_1, \ldots, \lambda_n^{-1}x_n\Big)\Big) \\ &< \lambda_1 \cdots \lambda_n\end{aligned}$$

if for all i, we have $\lambda_i^{-1}x_i \in U_i$. For each $i \in [n]$, we have U_i is an open neighborhood of $0 \in V_i$, so there exists some continuous seminorm σ_i and $\varepsilon_i > 0$ such that $U(0, \sigma_i, \varepsilon_i) \subseteq U_i$. If for every i, we have $\sigma_i\Big(\lambda_i^{-1}x_i\Big) < \varepsilon_i$, then

$$\tau(M(x_1, \ldots, x_n)) < \lambda_1 \cdots \lambda_n.$$

Rewriting, we have $\lambda_i > \frac{1}{\varepsilon_i}\sigma_i(x_i)$ for each such i, so that we have

$$\begin{aligned}\tau(M(x_1, \ldots, x_n)) &\leq \lambda_1 \cdots \lambda_n \\ &\leq \inf\left(\left\{\lambda_1 \cdots \lambda_n : \lambda_i > 0 \text{ and for all } i \in [n],\ \lambda_i > \frac{1}{\varepsilon_i}\sigma_i(x_i)\right\}\right) \\ &= \left(\frac{1}{\varepsilon_1}\sigma_1(x_1)\right) \cdots \left(\frac{1}{\varepsilon_n}\sigma_n(x_n)\right)\end{aligned}$$

Setting $\rho_i = \frac{1}{\varepsilon_i}\sigma_i$ yields the desired result.

Assume that (†) holds. Let $(x_1, \ldots, x_n), (y_1, \ldots, y_n) \in V_1 \times \cdots \times V_n$. We want to try to compare the difference between $M(x_1, \ldots, x_n)$ and $M(y_1, \ldots, y_n)$. From multilinearity, we have

$$M(x_1, \ldots, x_n) - M(y_1, \ldots, y_n) = \sum_{i=1}^{n} (M(y_1, \ldots, y_{i-1}, y_i, x_{i+1}, \ldots, x_n) - M(y_1, \ldots, y_{i-1}, x_i, x_{i+1}, \ldots, x_n))$$

$$= \sum_{i=1}^{n} M(y_1, \dots, y_{i-1}, y_i - x_i, x_{i+1}, \dots, x_n)$$

Let $Z \subseteq W$ be open containing $z = M(x_1, \dots, x_n)$, and write $z' = M(y_1, \dots, y_n)$. Then, there exists a continuous seminorm τ on W and $\varepsilon > 0$ such that $U(z, \tau, \varepsilon) \subseteq Z$.

Consider $\tau(z - z')$. Then, we have by multilinearity and the triangle inequality that

$$\tau(z - z') \leq \sum_{i=1}^{n} \tau(M(y_1, \dots, y_{i-1}, y_i - x_i, x_{i+1}, \dots, x_n)).$$

Using (†) gives continuous seminorms $\rho_1, \dots, \rho_n$ on $V_1, \dots, V_n$ respectively such that

$$\begin{aligned} &\leq \sum_{i=1}^{n} \rho_1(y_1) \cdots \rho_{i-1}(y_{i-1}) \rho_i(y_i - x_i) \rho_{i+1}(x_{i+1}) \cdots \rho_n(x_n) \\ &\leq n(C + \eta)^{n-1} \eta, \end{aligned}$$

where we assume that for all i, $y_i \in U(x_i, \rho_i, \eta)$, and $C = \max\{\rho_i(x_i) : i = 1, \dots, n\}$. By choosing η sufficiently small, $n(C + \eta)^{n-1} \eta < \varepsilon$, which yields

$$\begin{aligned} M\left(\prod_{i=1}^{n} U(x_i, \rho_i, \eta) \right) &\subseteq U(z, \tau, \varepsilon) \\ &\subseteq Z, \end{aligned}$$

so M is continuous. □

Theorem (Refinement of Previous Theorem)**:** Let $V_1, \dots, V_n$ and W be locally convex topological vector spaces, and let $\mathcal{N}$ be a defining collection of seminorms for the topology on W. Then, if $M : V_1 \times \cdots \times V_n \to W$ is a multilinear map, we have M is continuous if and only if for every $\tau \in \mathcal{N}$, there exist continuous seminorms $\rho_1, \dots, \rho_n$ on $V_1, \dots, V_n$ respectively such that for all $x = (x_1, \dots, x_n)$,

$$\tau(M(x)) \leq \prod_{i=1}^{n} \rho_i(x_i).$$

Proof. The forward direction follows from the fact that τ is a continuous seminorm, while the reverse direction follows from the fact that any continuous seminorm is dominated by a constant multiple of some family of seminorms in $\mathcal{N}$, and a constant multiple of the continuous seminorm ρ_i on V_i is also a continuous seminorm on V_i. □

Corollary: If $T : V \to W$ is a linear map of locally convex topological vector spaces $\mathcal{N}$ defining the topology on W, then T is continuous if and only if for all $\tau \in \mathcal{N}$ (or all continuous seminorms on W), there exists a continuous seminorm ρ on V such that for all $x \in V$, $\tau(T(x)) \leq \rho(x)$.

Proposition: Let V be a $\mathbb{K}$-vector space, and let $\mathcal{N}_1, \mathcal{N}_2 \subseteq \mathrm{SN}(V)$. Then, $\mathcal{T}(\mathcal{N}_1) \subseteq \mathcal{T}(\mathcal{N}_2)$ if and only if for all $\tau \in \mathcal{N}_1$, τ is a continuous seminorm for the locally convex topological vector space $(V, \mathcal{T}(\mathcal{N}_2))$.

Proof. We have that $\mathcal{T}(\mathcal{N}_1)$ is contained in $\mathcal{T}(\mathcal{N}_2)$ if and only if the identity map $\mathrm{id} : (V, \mathcal{T}(\mathcal{N}_2)) \to (V, \mathcal{T}(\mathcal{N}_1))$ is continuous. □

Remark: We recover the fact that $(V, \|\cdot\|_1)$ and $(V, \|\cdot\|_2)$ are equivalent as norms if and only if the identity map $\mathrm{id} : (V, \|\cdot\|_1) \to (V, \|\cdot\|_2)$ is a homeomorphism.

Normability and Metrizability

We will focus on Hausdorff locally convex topological vector spaces from now on. That is, the defining family of seminorms separates points.

Definition: Let $(V, \mathcal{T}(\mathcal{N}))$ be a Hausdorff locally convex topological vector space. We say V is normable if and only if there exists a norm $\|\cdot\|$ on V such that $\mathcal{T}(\mathcal{N}) = \mathcal{T}(\|\cdot\|)$.

Theorem: Let $(V, \mathcal{T}(\mathcal{N}))$ be a locally convex topological vector space. The following are equivalent:

(i) $(V, \mathcal{T}(\mathcal{N}))$ is normable;

(ii) there exists a finite subset F of $\mathcal{T}(\mathcal{N})$ such that $\mathcal{T}(\mathcal{N}) = \mathcal{T}(F)$;

Proof. We start by showing that (i) implies (ii). Let $\|\cdot\|$ be a norm such that $\mathcal{T}(\mathcal{N}) = \mathcal{T}(\|\cdot\|)$. Since $\mathcal{T}(\|\cdot\|) \subseteq \mathcal{T}(\mathcal{N})$, there exist $\rho_1, \ldots, \rho_n \in \mathcal{N}$ and some constant C such that

$$\|\cdot\| \leq C \max\{\rho_k : k = 1, \ldots, n\}.$$

Let $\tau \in \mathcal{N}$. Then, since $\mathcal{T}(\mathcal{N}) \subseteq \mathcal{T}(\|\cdot\|)$, it follows that there is some K such that $\tau \leq K\|\cdot\|$, whence

$$\tau \leq CK \max\{\rho_k : k = 1, \ldots, n\}$$

In particular, this means that the finite subset of seminorms $F = \{\rho_1, \ldots, \rho_n\}$ generates the topology $\mathcal{T}(\mathcal{N})$ since every seminorm in $\mathcal{N}$ is continuous with respect to $\mathcal{T}(F)$.

Next, we show that (ii) implies (i). Suppose there is a finite subset $F \subseteq \mathcal{N}$ with elements $\rho_1, \ldots, \rho_n$ such that $\mathcal{T}(\mathcal{N}) = \mathcal{T}(\mathcal{F})$. In particular, we have that for all seminorms $\tau \in \mathcal{N}$, we have

$$\tau \leq C \max\{\rho_k : k = 1, \ldots, n\}.$$

We claim that

$$\|\cdot\| = \max\{\rho_k : k = 1, \ldots, n\}$$

induces the topology on $\mathcal{N}$. This follows from the fact that every seminorm $\tau \in \mathcal{N}$ is continuous with respect to $\|\cdot\|$, so $\mathcal{T}(\mathcal{N}) \subseteq \mathcal{T}(\|\cdot\|) \subseteq \mathcal{T}(\mathcal{N})$, with the latter inclusion emerging from the fact that $\|\cdot\|$ is necessarily a continuous seminorm in $\mathcal{N}$ by the way it was defined. Since $(V, \mathcal{T}(\mathcal{N}))$ is Hausdorff, it follows that $\|\cdot\|$ separates points, so $\|\cdot\|$ is a norm. □

This allows us to prove that certain examples we discussed earlier are not normable spaces.

Example: Consider

$$V = \mathbb{R}^{\mathbb{N}}$$

with topology defined by seminorms $\rho_n = |x_n|$. Then, we observe that there are no finite subsets of the seminorms $\mathcal{N} = \{\rho_k : k \in \mathbb{N}\}$ that separate the points of V by virtue of the fact that if $F = \left\{\rho_{i_1}, \ldots, \rho_{i_j}\right\}$ is a finite subset, then the elementary vector e_{i_j+1} with 1 in position $\left(i_j + 1\right)$ is taken to 0 by any continuous seminorm on $\mathcal{T}(\mathcal{F})$, but e_{i_j+1} is not zero.

Next, we concern ourselves with metrizability of topological vector spaces.

Recall that (X, τ) is *first countable* if for any $x \in X$, the system of open neighborhoods $\mathcal{O}_x$ has a countable basis (i.e., a countable neighborhood basis). That is, there is some $\{U_n : n \geq 1\} \subseteq \mathcal{O}_x$ such that for any $Z \in \mathcal{O}_x$, there is some n such that $U_n \subseteq Z$. Metric spaces are always first countable by virtue of the Archimedean property.

Proposition: A (not necessarily locally convex) TVS $(V, \mathcal{T})$ is first countable if and only if 0 has a countable neighborhood basis.

Proof. Addition by a fixed vector $v \in V$ is a homeomorphism of V. □

Theorem: Let $(V, \mathcal{T}(\mathcal{N}))$ be a locally convex topological vector space. The following are equivalent:

(i) the space $(V, \mathcal{T}(\mathcal{N}))$ is metrizable;

(ii) the space $(V, \mathcal{T}(\mathcal{N}))$ is first countable;

(iii) there exists a countable subset $\mathcal{M} \subseteq \mathcal{N}$ such that $\mathcal{T}(\mathcal{M}) = \mathcal{T}(\mathcal{N})$.

Proof. We see that (i) implies (ii) since metrizable spaces are first countable.

Next, we show that (ii) implies (iii). Let $\{U_n : n \geq 0\}$ be a neighborhood basis for 0. Then, for any n, there exists finite $F_n \subseteq \mathcal{N}$ and $\varepsilon_n > 0$ such that

$$\bigcap_{\rho \in F_n} U(0, \rho, \varepsilon_n) \subseteq U_n.$$

Taking $\mathcal{M} = \bigcup_{n \geq 0} F_n$, this yields a countable subcollection of $\mathcal{N}$. We will show that $\mathcal{T}(\mathcal{N}) \subseteq \mathcal{T}(\mathcal{M})$. That is, we will have to show that for all $\tau \in \mathcal{N}$, τ is continuous with respect to $\mathcal{M}$.

We have $\tau^{-1}([0,1))$ is a neighborhood of 0 in $\mathcal{T}(\mathcal{N})$. This contains U_n for some $n \in \mathbb{N}$, and thus some finite subset F_n with

$$\begin{aligned} \bigcap_{\rho \in F_n} U(0, \rho, \varepsilon_n) &\subseteq U_n \\ &\subseteq \tau^{-1}([0,1)). \end{aligned}$$

In particular, this means that if $x \in V$ is such that for all $\rho \in F_n$, $\rho(x) < \varepsilon_n$, then $x \in \tau^{-1}([0,1))$, whence τ is continuous. Scaling allows one to apply the same method for any $x \in V$.

Finally, we will show that (iii) implies (i). Let $\mathcal{M} \subseteq \mathcal{N}$ be a countable subset with seminorms $\{\rho_n : n \geq 0\}$. Let

$$d(x,y) = \sum_{n=0}^{\infty} 2^{-n} \min\{\rho_n(x-y), 1\}.$$

Then, this is a distance metric and $\mathcal{T}(d) = \mathcal{T}(\mathcal{M})$. □

Definition: A locally convex topological vector space $(V, \mathcal{T})$. If $(V, \mathcal{T})$ is complete metrizable, then we call V a *Fréchet space*.

Example:

(i) Banach spaces are Fréchet spaces.

(ii) The space of rapidly decreasing sequences $s(\mathbb{N}, \mathbb{K})$ is a Fréchet space that is not normable. The distance metric is given by

$$d(x,y) = \sum_{k=0}^{\infty} 2^{-k} \min(1, \rho_k(x-y)).$$

(iii) The vector space $V = \mathbb{R}^{\mathbb{N}}$ with the product topology is a Fréchet space.

(iv) The space of finitely supported sequences $c_{00}(\mathbb{N}, \mathbb{K})$, henceforth denoted c_{00}, with the finest locally convex topology is a topological vector space that is not metrizable, let alone a Fréchet space. This follows from a diagonal argument on the assumption of countably many seminorms.

Consider the uncountable collection of seminorms $\mathcal{N}$ given by, for any $\omega \in [0,\infty)^{\mathbb{N}}$,

$$\rho_\omega(x) = \sum_{n=0}^{\infty} \omega_n |x_n|,$$

where we observe that each ρ_ω is distinct. It is possible to show that $\mathcal{T}(\mathcal{N}) = \mathcal{T}(\mathrm{SN}(c_{00}))$.

One direction follows directly from the definition. The reverse direction uses the fact that $\{e_n : n \geq 0\}$ forms an algebraic basis for $c_0 0$. If ρ is a seminorm, then we will define $\omega_n = \rho(e_n)$. For any $x \in c_{00}$,

then we may write

$$x = \sum_{n=0}^{N} x_n e_n,$$

satisfying

$$\begin{aligned}\rho(x) &= \rho\left(\sum_{n=0}^{N} x_n e_n\right) \\ &\leq \sum_{n=0}^{N} \omega_n |x_n| \\ &= \rho_\omega(x).\end{aligned}$$

Since x is arbitrary, it follows that every seminorm is continuous with respect to the seminorms of the family $\mathcal{N}$, whence the family $\mathcal{N}$ defines the topology on c_{00}.

In fact, we may assume that $\omega \in \mathbb{N}^{\mathbb{N}}$ by replacing $(\omega_n)_n \in [0,\infty)^{\mathbb{N}}$ with $(\lceil \omega_n \rceil)_n$.

Notable Examples

Example: Let $K \subseteq \mathbb{R}^d$ be a compact subset. Define

$$\mathcal{D}_K\left(\mathbb{R}^d\right) = C_K^\infty\left(\mathbb{R}^d\right)$$

to be the space of C^∞ with $\operatorname{supp}(f) \subseteq K$. Define seminorms for each $N \geq 0$ by

$$\rho_N(f) = \max_{\substack{\alpha \in \mathbb{N}^d \\ |\alpha| \leq N}} \sup_{x \in \mathbb{R}^d} \left|\partial^\alpha f(x)\right|.$$

Let $\mathcal{T} = \mathcal{T}(\{\rho_N : N \geq 0\})$. This is a countable collection of seminorms, so the space $C_K^\infty\left(\mathbb{R}^d\right)$ equipped with $\mathcal{T}$ is a Fréchet space with the metric

$$d(f,g) = \sum_{N \geq 0} 2^{-N} \min(1, \rho_N(f-g)).$$

Example: A function $f : \mathbb{R}^d \to \mathbb{C}$ is called a Schwartz function if $f \in C^\infty\left(\mathbb{R}^d\right)$ and for all $\alpha \in \mathbb{N}^d$ and all $k \in \mathbb{N}$,

$$\sup_{x \in \mathbb{R}^d} \langle x \rangle^k \left|\partial^\alpha f(x)\right| < \infty.$$

We label these seminorms by $\rho_{\alpha,k}(f)$. The space $\mathcal{S}\left(\mathbb{R}^d\right)$ of all such functions is known as the *Schwartz space* for $\mathbb{R}^d$.

The standard topology on $\mathcal{S}\left(\mathbb{R}^d\right)$ is given by $\mathcal{T} = \mathcal{T}(\mathcal{N})$, where $\mathcal{N}$ is the collection of all such $\rho_{\alpha,k}$. We may also write $\|\cdot\|_{\alpha,k}$ for $\rho_{\alpha,k}$

We can also consider the family of all such ρ_N with

$$\rho_N(f) = \max_{\substack{|\alpha| \leq N \\ k \leq N}} \rho_{\alpha,k}(f),$$

which yields the metric

$$d(f,g) = \sum_{N=0}^{\infty} 2^{-N} \min(1, \rho_N(f-g)).$$

This induces the same topology as the family of all the $\rho_{\alpha,k}$.

With this distance metric, the Schwartz space is complete. In particular, this means that $\mathcal{S}\left(\mathbb{R}^d\right)$ is an example of a Fréchet space. The Schwartz space is needed to define the Fourier transform (which we will discuss more heavily later). The Fourier transform is defined by

$$\hat{f}(\xi) = \int_{\mathbb{R}^d} e^{-ix\xi} f(x)\, d^d x.$$

When $f \in L^1$, we have that $\hat{f}$ is well-defined, but $\hat{f}$ is not necessarily in L^1. It turns out that the Fourier transform is an automorphism of the Schwartz space (over $\mathbb{C}$). Note that Schwartz functions are L^1 since they decay faster than functions of the form

$$u(x) = \langle x \rangle^{\lambda}$$

where $\lambda < d$. That such a $u(x)$ is integrable can be found by using the hyperspherical coordinates.

Example (A Non-Example)**:** The dual space of the Schwartz space, $\mathcal{S}\left(\mathbb{R}^d\right)^*$, is called the space of *tempered distributions*. We say $\mathcal{S}\left(\mathbb{R}^d\right)^* \subseteq C_c^\infty\left(\mathbb{R}^d\right)^*$, where the dual space $C_c^\infty\left(\mathbb{R}^d\right)^*$ is called the space of all distributions. We will discuss these more soon.

The space $C_c^\infty\left(\mathbb{R}^d\right)$ is the space of compactly supported C^∞ functions. In particular,

$$\begin{aligned} C_c^\infty\left(\mathbb{R}^d\right) &= \bigcup_{K \text{ compact}} C_K^\infty\left(\mathbb{R}^d\right) \\ &= \bigcup_{N \geq 1} C_{B(0,N)}^\infty\left(\mathbb{R}^d\right), \end{aligned}$$

where

$$B(0,N) = \left\{x \in \mathbb{R}^d : |x| \leq N\right\}$$

is the closed ball centered at 0.

Though $C_c^\infty\left(\mathbb{R}^d\right)$ is the increasing countable union of Fréchet spaces, it is not itself Fréchet; in fact, $C_c^\infty\left(\mathbb{R}^d\right)$ is not even metrizable. It is an example of an inductive limit of a family of Fréchet spaces.

Let

$$\iota_K \colon C_K^\infty\left(\mathbb{R}^d\right) \to C_c^\infty\left(\mathbb{R}^d\right)$$

denote the inclusion maps. We say a seminorm ρ on $C_c^\infty\left(\mathbb{R}^d\right)$ is compatible if and only if for every compact $K \subseteq \mathbb{R}^d$, $\rho \circ \iota_K$ is a continuous seminorm on $C_K^\infty\left(\mathbb{R}^d\right)$. The topology on $C_c^\infty\left(\mathbb{R}^d\right)$ is the topology generated by all compatible seminorms.

Definition: We say a sequence $(f_n)_n$ of functions on $U \subseteq \mathbb{R}^d$ converges locally uniformly to g if for all $x \in U$, there is an open subset $V \subseteq U$ containing x such that

$$\lim_{n \to \infty} \sup_{y \in V} \left|f_n(y) - g(y)\right| = 0.$$

Theorem: The sequence $(f_n)_n$ converges locally uniformly to g if and only if $(f_n)_n$ converges uniformly on compact subsets.

Proposition: Let $(f_n)_n$ be a sequence of C^1 functions on $\mathbb{R}^d$, and suppose f_n converges pointwise to $f \colon U \to \mathbb{C}$. Furthermore, assume that for all $i \in [d]$, $\partial_i f_n$ converges locally uniformly to some function $g_i \colon U \to \mathbb{C}$. Then, f is C^1 and $g_i = \partial_i f$ for all i.

Proof. Since each g_i is the local uniform limit of some continuous functions, each g_i is continuous. Fix $x \in U$.

Since $\partial_i f_n \to g_i$ locally uniformly, there is some $\varepsilon > 0$ such that

$$\sup_{y \in U(x,\varepsilon)} \left|\partial_i f_n(y) - g_i(y)\right| \xrightarrow{n\to\infty} 0.$$

For some $t \in \mathbb{R}$ with $0 < |t| < \varepsilon$, the fundamental theorem of calculus gives

$$f_n(x + te_i) - f_n(x) = t \int_0^1 \partial_i f_n(x + ste_i)\, ds.$$

Taking the limit as $n \to \infty$ and using dominated convergence gives

$$\begin{aligned} f(x + te_i) - f(x) &= \lim_{n\to\infty} t \int_0^1 \partial_i f_n(x + ste_i)\, ds \\ &= t \int_0^1 g_i(x + ste_i)\, ds. \end{aligned}$$

Therefore, we have

$$\begin{aligned} \left|\frac{f(x + te_i) - f(x)}{t} - g_i(x)\right| &= \left|\int_0^1 g_i(x + ste_i) - g(x)\, ds\right| \\ &\leq \sup_{0\leq s\leq 1} |g_i(x + ste_i) - g_i(x)| \\ &\to 0 \end{aligned}$$

by the continuity of g_i at x. Thus, $\partial_i f = g_i$, so f is C^1. □

Duality

Let $(V, \mathcal{T})$ be a topological vector space. The algebraic dual space V' is the space of all linear functionals $L\colon V \to \mathbb{C}$. The topological dual space, V^*, is the space of all continuous linear functionals $L\colon V \to \mathbb{C}$.

Definition: The weak* topology on V^* is the topology induced by the family of seminorms

$$\mathcal{N}_{\text{wk}^*} := \{(L \mapsto |L(x)|) : x \in V\}.$$

There is another locally convex topology on V^* that is the analogue of the norm topology in the case of a normed vector space. To understand this analogue, we must discuss boundedness.

Definition: Let $(V, \mathcal{T})$ be a topological vector space. We say A is bounded if for all open neighborhoods $U \in \mathcal{O}_0$ of 0, there is $\lambda > 0$ such that $\lambda A \subseteq U$.

Theorem: Let V be a normed vector space. Then, A is bounded in the above definition if and only if A is bounded in the usual sense.

Proposition: Let $(V, \mathcal{T})$ be a topological vector space, and let $A \subseteq V$. The following are equivalent:

(i) A is bounded;

(ii) for all $\rho \in \mathcal{N}$, $\sup_{x\in A} \rho(x) < \infty$;

(iii) for all continuous seminorms ρ, $\sup_{x\in A} \rho(x) < \infty$.

Proof. Assume (i); that is, suppose A is bounded, and let ρ be a defining seminorm. Then, there exists some $\lambda > 0$ such that $\lambda A \subseteq U(0, \rho, 1)$. That is, for all $x \in A$,

$$\lambda \rho(x) < 1,$$

or that $\rho(x) < \lambda^{-1}$. Therefore, $\sup_{x\in A} \rho(x) \leq \lambda^{-1}$, giving (ii).

Suppose (ii) holds. Let ρ be a continuous seminorm on $(V,\mathcal{T})$. Then, we may write $\rho(x) \leq C\max\{\rho_i(x) : i = 1,\dots,n\}$ for some family of defining seminorms ρ_i. This gives (iii).

Assume (iii) holds. Let $U \in \mathcal{O}_0$ be an open neighborhood of 0. Then, there exists some continuous seminorm ρ and some $\varepsilon > 0$ such that $U(0,\rho,\varepsilon) \subseteq U$. Notice that for any $x \in A$ and any $\lambda > 0$, we have

$$\begin{aligned}\rho(\lambda x) &= \lambda\rho(x)\\ &\leq \lambda \sup_{y\in A}\rho(y).\end{aligned}$$

Since $M = \sup_{y\in A}\rho(y)$ is finite by assumption, choosing λ sufficiently small such that $0 < \lambda < \varepsilon/M$ yields $\rho(\lambda x) < \varepsilon$ for all $x \in A$, or that $\lambda A \subseteq U(0,\rho,\varepsilon)$. □

Theorem: Let $(V,\mathcal{T})$ be a topological vector space. Then, $\mathcal{T}$ is a norm topology if and only if V contains a bounded nonempty open neighborhood of 0.

Definition: Let $(V,\mathcal{T})$ be a topological vector space. The *strong dual* is the topology on V^* defined by the family of all ρ_A with A bounded, where

$$\rho_A(L) = \sup_{x\in A}|L(x)|.$$

Definition: Let V and W be topological vector spaces with strong duals V^* and W^* respectively. Let $\varphi\colon V \to W$ be a continuous linear map. Then, the transpose is given by $\varphi^*\colon W^* \to V^*$ taking $\varphi^*(L) = L\circ\varphi$.

Proposition: The transpose of a continuous linear map is well-defined, linear, and continuous.

Proof. Our goal is to show that, given $L \in V^*$, we have that $\varphi^*(L) = L\circ\varphi$ is a continuous linear functional. This follows from the fact that L and φ are continuous. Similarly, the linearity of φ^* follows directly from definition.

To show that φ^* is continuous, the seminorm ρ_A on V^* for any $A \subseteq V$ bounded. Then, for any $L \in W^*$, we have

$$\begin{aligned}\rho_A(\varphi^*(L)) &= \sup_{x\in A}\big|L\circ\varphi(x)\big|\\ &= \sup_{y\in\varphi(A)}\big|L(y)\big|\\ &= \rho_{\varphi(A)}(L).\end{aligned}$$

We note that $\varphi(A)$ is itself bounded. □

Fourier Analysis

The next section of our coverage is focused on Fourier analysis.

Schwartz Distributions

Recall that the dual space $C_c^\infty\left(\mathbb{R}^d\right)^*$ of compactly supported C^∞ functions is called the space of *distributions*, while the dual space $\mathcal{S}\left(\mathbb{R}^d\right)^*$ is called the space of *tempered distributions*.

Given $\theta \in \mathcal{S}\left(\mathbb{R}^d\right)^*$ and $f \in \mathcal{S}\left(\mathbb{R}^d\right)$, we write $\langle f,\theta\rangle = \theta(f)$ (this will be held more generally for dual systems too). Since $C_c^\infty\left(\mathbb{R}^d\right) \subseteq \mathcal{S}\left(\mathbb{R}^d\right)$, we will define a canonical inclusion map $\varphi\colon C_c^\infty\left(\mathbb{R}^d\right) \to \mathcal{S}\left(\mathbb{R}^d\right)$.

Theorem: The inclusion φ is continuous with respect to the locally convex topology on each space.

Proof. Once again we use the seminorm criterion. Let $\alpha \in \mathbb{N}^d$ and $k \in \mathbb{N}$. Consider the seminorm $\|\cdot\|_{\alpha,k}$ on $\mathcal{S}$. We will show that there exists a continuous seminorm ρ on $C_c^\infty\left(\mathbb{R}^d\right)$ such that $\left\|\varphi(f)\right\|_{\alpha,k} \leq C\rho(f)$ for all $f \in C_c^\infty\left(\mathbb{R}^d\right)$.

It would suffice to show that for any compact $K \subseteq \mathbb{R}^d$, $\|\cdot\|_{\alpha,k} \circ \iota_K$ is a continuous seminorm on $C_c^\infty(K)$, where $\iota_K \colon C_c^\infty(K) \subseteq C_c^\infty\left(\mathbb{R}^d\right)$ is the inclusion. Indeed, we have

$$\begin{aligned}\|\iota_K(f)\|_{\alpha,k} &= \sup_{x\in\mathbb{R}^d} \langle x\rangle^k \left|\partial^\alpha \iota_K(f)(x)\right| \\ &= \sup_{x\in K} \langle x\rangle^k \left|\partial^\alpha f(x)\right| \\ &\leq \left(\sup_{x\in K}\langle x\rangle^k\right)\sup_{x\in K}\left|\partial^\alpha f(x)\right|.\end{aligned}$$

Defining $C = \sup_{x\in K}\langle x\rangle^k$, we have that $C < \infty$ by compactness, so we obtain our desired result. □

Theorem: The space of test functions, $C_c^\infty\left(\mathbb{R}^d\right)$, is sequentially dense in the Schwartz space.

Proof. We start by defining a smooth bump function $\varphi \colon \mathbb{R}^d \to [0,1]$ such that $\varphi(x) = 1$ for any $x \in [-1,1]^d$ and $\varphi(x) = 0$ for all $x \in \mathbb{R}^d \setminus (-2,2)^d$. Take $\rho_1, \rho_2 \colon \mathbb{R} \to \mathbb{R}$ by taking

$$\rho_1(x) = \begin{cases} 0 & x \leq 0 \\ e^{-1/x} & x > 0 \end{cases}$$
$$\rho_2(x) = \rho_1(x)\rho_1(1-x).$$

Then, ρ_1 is smooth, so ρ_2 is smooth as well. Moreover, $\rho_2(x) = 0$ for any $x \notin (0,1)$, and $\rho_2(x) \geq 0$ For any $x \in (0,1)$. Let

$$C = \int_0^1 \rho_2(x)\,dx$$
$$\rho_3(x) = \frac{1}{C}\int_0^x \rho_2(y)\,dy.$$

Then, $\rho_3(x) = 1$ for all $x \geq 1$, $\rho_3(x) = 0$ for all $x \leq 0$, and it is smooth in between. If we let ρ_4 be given by $\rho_4(x) = \rho_3(2-x)\rho_3(2+x)$, then for any $1 \leq x \leq 1$, we have $\rho_3(2-x) = \rho_3(2+x) = 1$, and $\rho_4(x) = 0$ for any $|x| \geq 2$, with ρ_4 otherwise smooth and bounded between 0 and 1. Finally, let $\varphi \colon \mathbb{R}^d \to [0,1]$ be given by

$$\varphi = \rho_4(x_1)\cdots\rho_4(x_d).$$

Let $f \in \mathcal{S}\left(\mathbb{R}^d\right)$. Let $\varphi_n(x) = \varphi\left(\frac{x}{n}\right)$, and let $f_n(x) = f(x)\varphi_n(x)$. Our goal is to show that $f_n \to f$ in the topology of $\mathcal{S}\left(\mathbb{R}^d\right)$. Let $\alpha \in \mathbb{N}^d$ and $k \in \mathbb{N}$. We must show that

$$\begin{aligned}\|f - f_n\|_{\alpha,k} &= \sup_{x\in\mathbb{R}^d}\langle x\rangle^k\left|\partial^\alpha f_n(x) - \partial^\alpha f(x)\right| \\ &\xrightarrow{n\to\infty} 0.\end{aligned}$$

Note that for any $x \in (-n,n)^d$, we have $|\partial^\alpha f_n(x) - \partial^\alpha f(x)| = 0$ since the truncation and the original function are equal on $[-n,n]^d$. Thus, we only need to consider $x \notin (-n,n)^d$. By the Leibniz rule, we have

$$\begin{aligned}\partial^\alpha f_n(x) &= \partial^\alpha(\varphi_n(x)f(x)) \\ &= \sum_{\substack{\beta,\gamma\in\mathbb{N}^d \\ \beta+\gamma=\alpha}} \frac{\alpha!}{\beta!\gamma!}\partial^\beta\varphi_n(x)\partial^\gamma f(x).\end{aligned}$$

Applying chain rule gives

$$\partial^\alpha f_n(x) = \sum_{\substack{\beta,\gamma\in\mathbb{N}^d \\ \beta+\gamma=\alpha}} \frac{\alpha!}{\beta!\gamma!} n^{-|\beta|}\left(\partial^\beta\varphi\right)\left(\frac{x}{n}\right)\partial^\gamma f(x).$$

Since the $\gamma = \alpha$ term is eliminated, we get

$$\partial^\alpha f_n(x) - \varphi_n(x)\partial^\alpha f(x) = \sum_{\substack{\beta,\gamma\in\mathbb{N}^d \\ \beta+\gamma=\alpha, |\beta|\geq 1}} \frac{\alpha!}{\beta!\gamma!} n^{-|\beta|}\Big(\partial^\beta \varphi\Big)\Big(\frac{x}{n}\Big)\partial^\gamma f(x)$$

Therefore, we obtain

$$\Big|\partial^\alpha f_n(x) - \varphi_n(x)\partial^\alpha f(x)\Big| \leq \sum_{\substack{\beta,\gamma\in\mathbb{N}^d \\ \beta+\gamma=\alpha, |\beta|\geq 1}} \frac{\alpha!}{\beta!\gamma!} \big\|\varphi\big\|_{\beta,0} \langle x\rangle^{-k-1} \|f\|_{\gamma,k+1}.$$

Choose some constant C_1 greater than or equal to the right hand side, so that we have

$$\langle x\rangle^k \Big|\partial^\alpha f_n(x) - \varphi_n(x)\partial^\alpha f(x)\Big| \leq C_1 \langle x\rangle^{-1}.$$

Therefore, by the triangle inequality, we have

$$\langle x\rangle^k \Big|\partial^\alpha f_n(x) - \partial^\alpha f(x)\Big| \leq C_1\langle x\rangle^{-1} + \langle x\rangle^k \Big|\varphi_n(x)\partial^\alpha f(x) - \partial^\alpha f(x)\Big|.$$

Since $0 \leq \varphi_n(x) \leq 1$, we have

$$\begin{aligned}\langle x\rangle^k \Big|\varphi_n(x)\partial^\alpha f(x) - \partial^\alpha f(x)\Big| &\leq \langle x\rangle^k \Big(\langle x\rangle^{-k-1}\|f\|_{\alpha,k+1}\Big)\Big|\varphi_n(x) - 1\Big| \\ &\leq \langle x\rangle^{-1}\|f\|_{\alpha,k+1},\end{aligned}$$

so that

$$\langle x\rangle^k \Big|\partial^\alpha f_n(x) - \partial^\alpha f(x)\Big| \leq \frac{C + \|f\|_{\alpha,k+1}}{\langle x\rangle}.$$

Since for any $x \notin (-n,n)^d$, we have $\langle x\rangle \geq |x| \geq n$, it follows that

$$\begin{aligned}\|f_n - f\|_{\alpha,k} &= \sup_{x\in\mathbb{R}^d} \langle x\rangle^k \Big|\partial^\alpha f_n(x) - \partial^\alpha f(x)\Big| \\ &\leq \frac{C_1 + \|f\|_{\alpha,k}}{\langle x\rangle} \\ &\leq \frac{C_1 + \|f\|_{\alpha,k+1}}{n} \\ &\xrightarrow{n\to\infty} 0.\end{aligned}$$

□

Corollary: The space of tempered distributions, $\mathcal{S}\big(\mathbb{R}^d\big)^*$ is contained in the space of distributions $C_c^\infty\big(\mathbb{R}^d\big)^*$.

Proof. We have shown that the inclusion map $\varphi\colon C_c^\infty\big(\mathbb{R}^d\big) \to \mathcal{S}\big(\mathbb{R}^d\big)$ is linear and continuous, so the transpose $\varphi^*\colon \mathcal{S}\big(\mathbb{R}^d\big)^* \to C_c^\infty\big(\mathbb{R}^d\big)^*$ is linear and continuous. Our goal is to show the injectivity of the transpose map.

Yet, this follows from the fact that if $L \in \mathcal{S}\big(\mathbb{R}^d\big)^*$ is such that $\varphi^*(L) = 0$, then $L(f) = 0$ for all $f \in C_c^\infty\big(\mathbb{R}^d\big)$, but since L is continuous and $C_c^\infty\big(\mathbb{R}^d\big) \subseteq \mathcal{S}\big(\mathbb{R}^d\big)$ is sequentially dense, we have that $L = 0$ so φ^* is injective. □

Classical Fourier Transform

We will assume that $\mathbb{K} = \mathbb{C}$.

Definition: Let

$$\begin{aligned}\mathcal{F}\colon \mathcal{S}\left(\mathbb{R}^d,\mathbb{C}\right) &\to \mathcal{S}\left(\mathbb{R}^d,\mathbb{C}\right)\\ f &\mapsto \hat{f} = \mathcal{F}[f]\end{aligned}$$

be defined by

$$\hat{f}(\xi) := \int_{\mathbb{R}^d} e^{-i\langle x,\xi\rangle} f(x)\, d^d x.$$

Remark: We will let

$$\mathcal{G}\colon \mathcal{S}\left(\mathbb{R}^d,\mathbb{C}\right) \to \mathcal{S}\left(\mathbb{R}^d,\mathbb{C}\right)$$

be defined by

$$\mathcal{G}\left[\hat{f}\right](x) = \int_{\mathbb{R}^d} e^{i\langle \xi,x\rangle} \hat{f}(\xi)\, d^d \xi.$$

Theorem: The map $\mathcal{F}$ is well-defined, linear, and continuous.

Proof. We recall the seminorms

$$\|f\|_{\alpha,k} = \sup_{x\in\mathbb{R}^d} \langle x\rangle^k \left|\partial^\alpha f(x)\right|.$$

We want to verify that $\mathcal{F}$ is well-defined. We note that

$$\begin{aligned}\int_{\mathbb{R}^d} \left|e^{-i\langle x,\xi\rangle} f(x)\right| d^d x &= \int_{\mathbb{R}^d} \left|\langle x\rangle^{-(d+1)} \langle x\rangle^{d+1} f(x)\right| d^d x\\ &\leq \|f\|_{0,d+1} \int_{\mathbb{R}^d} \langle x\rangle^{-(d+1)}\, d^d x\\ &< \infty.\end{aligned}$$

Our next job is to show that $\hat{f}$ is C^∞. By dominated convergence, we have that

$$\begin{aligned}\partial_j \hat{f}(\xi) &= \int_{\mathbb{R}^d} \frac{\partial}{\partial \xi_j}\left(e^{-i\langle x,\xi\rangle} f(x)\right) d^d x\\ &= \int_{\mathbb{R}^d} \left(-ix_j\right) e^{-i\langle x,\xi\rangle} f(x)\, d^d x.\end{aligned}$$

Dominated convergence applies since applying the derivative to the integrand yields the upper bound given by

$$\begin{aligned}\left|\left(-ix_j\right) e^{-i\langle x,\xi\rangle} f(x)\right| &\leq \sqrt{1 + x_1^2 + \cdots + x_n^2}|f(x)|\\ &= \langle x\rangle |f(x)|\\ &=: g(x),\end{aligned}$$

which is integrable since f is a Schwartz function.

Inductively, we have

$$\left|\frac{\partial^n}{\partial \xi_{j_1}\cdots\partial\xi_{j_1}} e^{-i\langle x,\xi\rangle} f(x)\right| \leq \langle x\rangle^n |f(x)|.$$

In general, for any multi-index $\alpha \in \mathbb{N}^d$, we have

$$\partial^\alpha \hat{f}(\xi) = \int_{\mathbb{R}^d} (-i)^{|\alpha|} x^\alpha e^{-i\langle x,\xi\rangle} f(x)\, d^d x.$$

Therefore, $\hat{f}$ is C^∞.

Our next goal is to extract the decay rate so that $\hat{f}$ is in Schwartz space. Let $\alpha, \beta \in \mathbb{N}^d$, and consider

$$\begin{aligned}\xi^\beta \partial_\xi^\alpha \hat{f}(\xi) &= \int_{\mathbb{R}^d} (-i)^{|\alpha|} x^\alpha \xi^\beta e^{-i\langle x,\xi\rangle} f(x)\, d^d x \\ &= i^{|\beta|-|\alpha|} \int_{\mathbb{R}^d} \partial_x^\beta \left(e^{-i\langle x,\xi\rangle}\right) x^\alpha f(x)\, d^d x,\end{aligned}$$

where we note that $\partial_x^\beta \left(e^{-i\langle x,\xi\rangle}\right) = i^{-|\beta|}\xi^\beta e^{-i\langle x,\xi\rangle}$.

Integrating by parts and the Leibniz rule yield

$$\begin{aligned}\xi^\beta \partial_\xi^\alpha \hat{f}(\xi) &= (-i)^{|\alpha|+|\beta|} \int_{\mathbb{R}^d} e^{-i\langle x,\xi\rangle} \partial_x^\beta (x^\alpha f(x))\, d^d x \\ &= (-i)^{|\alpha|+|\beta|} \int e^{-i\langle x,\xi\rangle} \sum_{\substack{0\le\gamma\le\beta \\ \gamma\in\mathbb{N}^d}} \frac{\beta!}{\gamma!(\beta-\gamma)!} \partial_x^\gamma (x^\alpha) \partial_x^{\beta-\gamma}(f(x))\, d^d x \\ &= (-i)^{|\alpha|+|\beta|} \sum_{\gamma\in\mathbb{N}^d} \chi_{\{\gamma\le\min(\alpha,\beta)\}} \frac{\alpha!\beta!}{\gamma!(\alpha-\gamma)!(\beta-\gamma)!} \int e^{-i\langle x,\xi\rangle}\, d^d x.\end{aligned}$$

Estimating yields

$$\left|\xi^\beta \partial_\xi^\alpha \hat{f}(\xi)\right| \le \sum_{\gamma\in\mathbb{N}^d} \chi_{\{\gamma\le\min(\alpha,\beta)\}} \frac{\alpha!\beta!}{\gamma!(\alpha-\gamma)!(\beta-\gamma)!} \left(\int_{\mathbb{R}^d} \langle x\rangle^{-(d+1)}\, d^d x\right) \|f\|_{\beta-\gamma,|\alpha|-|\gamma|+d+1}.$$

In particular, the right-hand side is a finite constant, so that

$$\begin{aligned}\sup_{\xi\in\mathbb{R}^d} \left|\langle\xi\rangle^{2m} \partial_\xi^\alpha \hat{f}(\xi)\right| &= \sup_{\xi\in\mathbb{R}^d} \left(1+\xi_1^2+\cdots+\xi_d^2\right)^m \left|\partial_\xi^\alpha \hat{f}(\xi)\right| \\ &< \infty.\end{aligned}$$

Thus, $\hat{f} \in \mathcal{S}\left(\mathbb{R}^d\right)$. Furthermore, this estimate yields that $\mathcal{F}$ is continuous, as any defining seminorm applied to $\hat{f}$ is bounded by a constant multiple of a seminorm applied to f.

Similarly, the map $\mathcal{G}$ simply substitutes i for $-i$, so that $\mathcal{G}$ is also continuous. □

The most important Fourier transform that we will compute is the Gaussian. That is, $f(x) = e^{-|x|^2/2}$.

Theorem: If $f(x) = e^{-|x|^2/2}$ over $\mathbb{R}^d$, we have $\hat{f}(\xi) = (2\pi)^{d/2} e^{-\xi^2/2}$.

Proof. Notice that $f(x) \in \mathcal{S}\left(\mathbb{R}^d\right)$. Fubini's theorem yields

$$\begin{aligned}\hat{f}(\xi) &= \int_{\mathbb{R}^d} e^{-i\langle x,\xi\rangle} e^{-|x|^2/2}\, d^d x \\ &= \int_{\mathbb{R}^d} \prod_{j=1}^d e^{-i\xi_j x_j - \left(x_j^2/2\right)}\, d^d x \\ &= \prod_{j=1}^d \int_{\mathbb{R}} e^{-|x|^2/2 - i\langle\xi,x\rangle}\, dx.\end{aligned}$$

Therefore, we reduce to the one-dimensional case. One can complete the square, yielding

$$\begin{aligned}\int_{-\infty}^{\infty} e^{-|x|^2/2-i\langle \xi,x\rangle}\,dx &= e^{-\xi^2/2}\int_{-\infty}^{\infty} e^{-\frac{1}{2}(x+i\xi)^2}\,dx \\ &= e^{-\xi^2/2}\lim_{A\to\infty}\int_{-A}^{A} e^{-\frac{1}{2}(x+i\xi)^2}\,dx\end{aligned}$$

Translating by $-i\xi$ and using the residue theorem on the resulting rectangular contour yields that the integral is the same as the original Gaussian integral.

The second method follows by a Taylor expansion. By dominated convergence, we have

$$\begin{aligned}\int_{-\infty}^{\infty} e^{-|x|^2/2-i\langle \xi,x\rangle}\,dx &= \int_{-\infty}^{\infty} x^n e^{-|x|^2/2}\sum_{n=0}^{\infty}\frac{(i|\xi|)^n}{n!}\,dx \\ &= \sum_{n=0}^{\infty}\frac{(i|\xi|)^n}{n!}\int_{-\infty}^{\infty} x^n e^{-|x|^2/2}\,dx.\end{aligned}$$

If X is a random variable with standard normal distribution, we note that the integral evaluates to

$$\int_{-\infty}^{\infty} x^n e^{-|x|^2}\,dx = (2\pi)^{1/2}\mathbb{E}[X^n],$$

while

$$\mathbb{E}[X^n] = \begin{cases} 0 & n \text{ is odd} \\ \frac{(2n)!}{2^n n!} & n \text{ is even.}\end{cases}$$

In particular, we have that

$$\begin{aligned}\int_{-\infty}^{\infty} e^{-|x|^2/2-i\langle \xi,x\rangle}\,dx &= (2\pi)^{1/2}\sum_{k=0}^{\infty}\frac{(-i)^{2k}\xi^{2k}}{(2k)!}\frac{(2k)!}{2^k k!} \\ &= (2\pi)^{1/2}e^{-\xi^2/2}.\end{aligned}$$

□

Our goal is to prove the Fourier Inversion Theorem.

Theorem (Fourier Inversion Theorem)**:** Let $\mathcal{F}$ and $\mathcal{G}$ be defined by

$$\begin{aligned}\mathcal{F}[f](\xi) &= \int_{\mathbb{R}^d} e^{-i\langle x,\xi\rangle} f(x)\,d^d x \\ \mathcal{G}\left[\hat{f}\right](x) &= \int_{\mathbb{R}^d} e^{i\langle x,\xi\rangle}\hat{f}(\xi)\,d^d\xi.\end{aligned}$$

Then, $\mathcal{G}\circ\mathcal{F} = \mathcal{F}\circ\mathcal{G} = (2\pi)^d$.

In particular, we obtain the following corollary.

Corollary: The map $\mathcal{F}$ is a topological vector space isomorphism from $\mathcal{S}\left(\mathbb{R}^d,\mathbb{C}\right)$ to itself, with inverse given by

$$\mathcal{F}^{-1}[f](x) = \frac{1}{(2\pi)^d}\int_{\mathbb{R}^d} e^{i\langle x,\xi\rangle} f(\xi)\,d^d\xi.$$

We will write $\check{f}(x)$ for $\mathcal{F}^{-1}[f](x)$.

To prove the Fourier Inversion Theorem, we will need the following proposition.

Proposition: We have

$$\lim_{\varepsilon \searrow 0} (2\pi\varepsilon)^{-d/2} e^{-\frac{|y-x|^2}{2\varepsilon}} = \delta_x^d(y)$$

in $\mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right)^*$ with the strong topology. Here, δ_x^d is the evaluation map implemented by the duality pairing

$$\left\langle f, \delta_x^d \right\rangle = \int_{\mathbb{R}^d} f(y)\delta_x^d(y)\, d^d y.$$

Remark: The duality pairing for $f \in \mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right)$ and $\varphi \in \mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right)^*$ written as

$$\langle f, \varphi \rangle = \int f(y)\varphi(y)\, dy$$

is not really an integral. However, physicists like writing it this way.

Proof of Proposition. Let $\varphi_{x,\varepsilon} \in \mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right)$ be defined by

$$\varphi_{x,\varepsilon}(y) = (2\pi\varepsilon)^{-d/2} e^{-\frac{|y-x|^2}{2\varepsilon}}.$$

There is a canonical injection $\iota \colon \mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right) \to \mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right)^*$.

Let $I = [0, \infty)$ be directed by taking $\varepsilon_1 \preceq \varepsilon_2$ if and only if $\varepsilon_2 \leq \varepsilon_1$. Then, our goal is to show that the net $(\iota(\varphi_{x,\varepsilon}))_{\varepsilon \in I}$ converges to δ_x^d in the strong topology.

We want to show that, for all bounded subsets $A \subseteq \mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right)$, one has

$$\begin{aligned} 0 &= \lim_{\varepsilon \searrow 0} \left\| \iota_{\varphi_{x,\varepsilon}} - \delta_x^d \right\|_A \\ &= \sup_{f \in A} \left| \left\langle f, \iota(\varphi_{x,\varepsilon}) - \delta_x^d \right\rangle \right|. \end{aligned}$$

We see that

$$\begin{aligned} \left\langle f, \iota(\varphi_{x,\varepsilon}) - \delta_x^d \right\rangle &= \int_{\mathbb{R}^d} \varphi_{x,\varepsilon}(y) f(y)\, d^d y - f(x) \\ &= \int_{\mathbb{R}^d} \varphi_{x,\varepsilon}(y)(f(y) - f(x))\, d^d y, \end{aligned}$$

where we note that $\varphi_{x,\varepsilon}$ was specifically chosen to have integral 1 over $\mathbb{R}^d$. Using the Fundamental Theorem of Calculus, we have

$$\begin{aligned} &= \int_{\mathbb{R}^d} \varphi_{x,\varepsilon}(y) \int_0^1 \langle y - x, \nabla f(x + t(y-x)) \rangle\, dt\, d^d y \\ &= \int_{\mathbb{R}^d} (2\pi\varepsilon)^{-d/2} e^{-\frac{|y-x|^2}{2}} \int_0^1 \langle (y-x), \nabla f(x + t(y-x)) \rangle\, dt\, d^d y \\ &= \sum_{j=1}^d \int_{\mathbb{R}^d} (2\pi\varepsilon)^{-d/2} e^{-\frac{|y-x|^2}{2\varepsilon}} \int_0^1 \left\langle y_j - x_j, \partial_j f(x + t(y-x)) \right\rangle dt\, d^d y. \end{aligned}$$

Estimating the inner integral yields

$$\left| \left\langle f, \iota(\varphi_{x,\varepsilon}) - \partial_x^d \right\rangle \right| \leq \sum_{j=1}^d \|f\|_{e_j,0} \int_{\mathbb{R}^d} \left| y_j - x_j \right| (2\pi\varepsilon)^{-d/2} e^{-\frac{|y-x|^2}{2\varepsilon}}\, d^d y$$

$$\leq d \max_{1\leq j\leq d} \int_{\mathbb{R}^d} (2\pi\varepsilon)^{-d/2} e^{-\frac{|y-x|^2}{2\varepsilon}} \left|y-x\right| d^d y.$$

Substituting $u = \frac{y-x}{\sqrt{\varepsilon}}$ yields $y = x + \sqrt{\varepsilon}u$, so that

$$\leq d\left(\max_{1\leq j\leq d} \sup_{f\in A} \|f\|_{e_j,0}\right)\sqrt{\varepsilon}\left(\int_{\mathbb{R}^d} (2\pi)^{-d/2} e^{-\frac{|u|^2}{2}} |u|\, d^d u\right).$$

Since the integral is finite, we obtain that $\left\|\iota(\varphi_{x,\varepsilon}) - \delta_x^d\right\|_A \leq C\sqrt{\varepsilon}$. □

Remark: The functions of the form $(2\pi\varepsilon)^{-d/2} e^{-\frac{|y-x|^2}{2\varepsilon}}$ are a special class of functions known as mollifiers. We will now prove the Fourier Inversion Theorem.

Proof of the Fourier Inversion Theorem. Let $f \in \mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right)$. We compute

$$\mathcal{G}\circ\mathcal{F}[f](x) = \int_{\mathbb{R}^d} e^{i\langle x,\xi\rangle} \hat{f}(\xi)\, d^d\xi.$$

Since $\hat{f}$ is a Schwartz function, dominated convergence gives

$$\begin{aligned}
&= \lim_{\varepsilon\searrow 0} \int e^{i\langle x,\xi\rangle - \frac{\varepsilon}{2}|\xi|^2} \hat{f}(\xi)\, d\xi \\
&= \int_{\mathbb{R}^d}\left(\int_{\mathbb{R}^d} e^{-i\langle x,\xi\rangle - \frac{\varepsilon}{2}|\xi|^2} e^{-i\langle y,\xi\rangle} f(y)\, d^d y\right) d^d\xi
\end{aligned}$$

Making use of the factor of $e^{-\frac{\varepsilon}{2}|\xi|^2}$ allows us to use Fubini's theorem to obtain

$$= \int_{\mathbb{R}^d}\left(\int_{\mathbb{R}^d} e^{i\langle x,\xi\rangle - \frac{\varepsilon}{2}|\xi|^2} e^{-i\langle y,\xi\rangle}\, d^d\xi\right) f(y)\, d^d y.$$

Evaluating the inner integral, we substitute $\eta = \sqrt{\varepsilon}\xi$ to obtain

$$\begin{aligned}
\int_{\mathbb{R}^d} e^{-\langle y-x,\xi\rangle} e^{-\frac{\varepsilon}{2}|\xi|^2}\, d^d\xi &= \varepsilon^{-d/2} \int_{\mathbb{R}^d} e^{-i\left(\left\langle \frac{y-x}{\sqrt{\varepsilon}},\eta\right\rangle\right)} e^{-\frac{|\eta|^2}{2}}\, d^d\eta \\
&= \varepsilon^{-d/2}\mathcal{F}\left[\eta\mapsto e^{-\frac{|\eta|^2}{2}}\right]\left(\frac{y-x}{\sqrt{\varepsilon}}\right) \\
&= (\varepsilon)^{-d/2}(2\pi)^{d/2} e^{-\frac{|y-x|^2}{2\varepsilon}} \\
&= (2\pi)^d\left((2\pi)^{-d/2} e^{-\frac{|y-x|^2}{2\varepsilon}}\right).
\end{aligned}$$

Rewriting as $\varphi_{x,\varepsilon}$, we have

$$\begin{aligned}
\mathcal{G}\circ\mathcal{F}[f](x) &= (2\pi)^d \lim_{\varepsilon\searrow 0} \int_{\mathbb{R}^d} f(y)\varphi_{x,\varepsilon}(y)\, d^d y \\
&= (2\pi)^d \delta_x^d(f) \\
&= (2\pi)^d f(x).
\end{aligned}$$

□

Theorem (Plancherel's Theorem)**:** For any $f, g \in \mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right)$, one has

$$\int_{\mathbb{R}^d} \overline{\hat{f}(\xi)}\hat{g}(\xi)\, d^d\xi = (2\pi)^d \int \overline{f(x)}g(x)\, d^d x.$$

Proof. The left-hand side can be written as follows

$$\int_{\mathbb{R}^d} \overline{\left(\int_{\mathbb{R}^d} e^{-i\langle x,\xi\rangle} f(x)\, d^d x\right)}\hat{g}(\xi)\, d^d\xi = \int_{\mathbb{R}^d}\int_{\mathbb{R}^d} e^{i\langle x,\xi\rangle}\overline{f(x)}\hat{g}(\xi)\, d^d\xi\, d^d x \tag{$*$}$$

This follows from the estimate with Tonelli's Theorem

$$\int_{\mathbb{R}^d}\int_{\mathbb{R}^d}\left|e^{i\langle x,\xi\rangle}\overline{f(x)}\hat{g}(\xi)\right| d^d x\, d^d\xi = \left(\int_{\mathbb{R}^d}|f(x)|\, d^d x\right)\left(\int_{\mathbb{R}^d}|\hat{g}(\xi)|\, d^d\xi\right)$$

and using the fact that functions on the Schwartz space are integrable. Yet again using Fubini's theorem on $(*)$, we have

$$\begin{aligned}\int_{\mathbb{R}^d}\int_{\mathbb{R}^d} e^{i\langle x,\xi\rangle}\overline{f(x)}\hat{g}(\xi)\, d^d\xi\, d^d x &= \int_{\mathbb{R}^d}\overline{f(x)}\left(\int_{\mathbb{R}^d} e^{i\langle x,\xi\rangle}\hat{g}(\xi)\, d^d\xi\right) d^d x \\ &= \int_{\mathbb{R}^d}\overline{f(x)}\mathcal{G}(\mathcal{F}[g])(x)\, d^d x \\ &= (2\pi)^d\int_{\mathbb{R}^d}\overline{f(x)}g(x)\, d^d x.\end{aligned}$$

□

We note that one can show that $\mathcal{U} = (2\pi)^{-d/2}\mathcal{F}$ is an isometry on the Schwartz space $\mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right)$. Since $\mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right)$ is dense in $L^2\left(\mathbb{R}^d\right)$, continuity gives that there is a unique extension $\widetilde{\mathcal{U}}\colon L^2\left(\mathbb{R}^d\right) \to L^2\left(\mathbb{R}^d\right)$. The map $\widetilde{\mathcal{U}}$ is the definition of the Fourier transform on $L^2\left(\mathbb{R}^d\right)$.

Convolution and Properties of the Fourier Transform on $\mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right)$

Definition: We define the space $\mathcal{O}_M\left(\mathbb{R}^d, \mathbb{K}\right)$ to consist of all functions $f\colon \mathbb{R}^d \to \mathbb{K}$ such that f is C^∞ and for every $\alpha \in \mathbb{N}^d$, there exists $k \in \mathbb{N}$ and a constant $C \geq 0$ such that for all $x \in \mathbb{R}^d$, one has

$$\left|\partial^\alpha f(x)\right| \leq C\langle x\rangle^k.$$

These are known as the smooth functions of tempered growth.

Proposition: Let $g \in \mathcal{O}_M$. Then, for all $f \in \mathcal{S}\left(\mathbb{R}^d\right)$, $fg \in \mathcal{S}\left(\mathbb{R}^d\right)$, and one has that the map $f \mapsto fg$ is continuous.

Proof. We will assume that $|\partial^\alpha g(x)| \leq C\langle x\rangle^{k_\alpha}$.

Now, we have

$$\partial^\alpha(fg) = \sum_{\substack{\beta\in\mathbb{N}^d \\ \beta\leq\alpha}} \frac{\alpha!}{\beta!(\alpha-\beta)!}\partial^{\alpha-\beta}f\,\partial^\beta f.$$

So,

$$\langle x\rangle^k\left|\partial^\alpha(fg)(x)\right| \leq \langle k\rangle \sum_{\substack{\beta\in\mathbb{N}^d \\ \beta\leq\alpha}} \frac{\alpha!}{\beta!(\alpha-\beta!)}\left|\partial^{\alpha-\beta}f(x)\right| C_\beta\langle x\rangle^{k_\beta}$$

$$\leq \sum_{\substack{\beta\in\mathbb{N}^d \\ \beta\leq\alpha}} \frac{\alpha!}{\beta!(\alpha-\beta)!} C_\beta \|f\|_{\alpha-\beta,k+k_\beta}$$
$$< \infty.$$

So, one has that $\|fg\|_{\alpha,k} < \infty$ for all $\alpha \in \mathbb{N}^d$ and all $k \in \mathbb{N}$. The seminorm criterion also gives continuity of the map $f \mapsto fg$. □

Remark: There is a converse to this theorem. Namely, if $g\colon \mathbb{R}^d \to \mathbb{K}$ is a function such that $fg \in \mathcal{S}\left(\mathbb{R}^d\right)$ for all $f \in \mathcal{S}\left(\mathbb{R}^d\right)$.

Bumpfunctionology (for lack of a better term) gives smoothness of such a g, but showing the polynomial bound requires a bit more work.

Example: Monomials $x \mapsto x^\alpha$ satisfy

$$\partial^\beta x^\alpha = \chi_{\{\beta\leq\alpha\}} \frac{\alpha!}{(\alpha-\beta)!} x^{\alpha-\beta},$$

so since $\left|x_j\right| \leq |x| \leq \langle x\rangle$, one has

$$\left|\partial^\beta x^\alpha\right| \leq C_\beta \langle x\rangle^{k_\beta},$$

where $k_\beta = \max\left(0, \left|\alpha-\beta\right|\right)$, and $C_\beta = \chi_{\{\beta\leq\alpha\}} \frac{\alpha!}{(\alpha-\beta)!}$.

We use

$$m_\alpha\colon \mathcal{S}\left(\mathbb{R}^d, \mathbb{K}\right) \to \mathcal{S}\left(\mathbb{R}^d, \mathbb{K}\right)$$
$$f \mapsto x^\alpha f.$$

Proposition: The operator $\partial^\alpha\colon \mathcal{S}\left(\mathbb{R}^d, \mathbb{K}\right) \to \mathcal{S}\left(\mathbb{R}^d, \mathbb{K}\right)$ taking $f \mapsto \partial^\alpha f$ is linear and continuous.

Proof. To be written. □

Theorem (Intertwining Property of Fourier Transform)**:** As continuous linear operators on $\mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right)$, one has

$$\partial^\alpha \mathcal{F} = (-i)^{|\alpha|} \mathcal{F} \circ m_\alpha$$
$$m_\alpha \circ \mathcal{F} = (-i)^{|\alpha|} \mathcal{F} \circ \partial^\alpha$$

Proof. Using dominated convergence, we may bring the derivative inside the integral, so that

$$\begin{aligned}
\partial_\xi^\alpha \hat{f}(\xi) &= \partial_\xi^\alpha \int_{\mathbb{R}^d} e^{-i\langle x,\xi\rangle} f(x)\, d^d x \\
&= \int_{\mathbb{R}^d} \partial_\xi^\alpha \left(e^{-i\langle x,\xi\rangle}\right) f(x)\, d^d x \\
&= \int_{\mathbb{R}^d} (-ix)^{|\alpha|} e^{-i\langle x,\xi\rangle} f(x)\, d^d x \\
&= (-i)^{|\alpha|} \int_{\mathbb{R}^d} e^{-i\langle x,\xi\rangle} x^\alpha f(x)\, d^d x \\
&= (-i)^{|\alpha|} (\mathcal{F} \circ m_\alpha[f])(x).
\end{aligned}$$

For the other direction, integration by parts yields

$$\xi^\alpha \hat{f}(\xi) = \int_{\mathbb{R}^d} \xi^\alpha e^{-i\langle x,\xi\rangle} f(x)\, d^d x$$

$$
\begin{aligned}
&= \int_{\mathbb{R}^d} i^{|\alpha|} \partial_x^\alpha \left(e^{-i\langle x,\xi\rangle} \right) f(x)\, d^d x \\
&= (-i)^{|\alpha|} \int_{\mathbb{R}^d} e^{-i\langle x,\xi\rangle} \partial_x^\alpha (f(x))\, d^d x \\
&= (-i)^{|\alpha|} (\mathcal{F} \circ \partial^\alpha [f])(x)
\end{aligned}
$$

□

Proposition: Pointwise multiplication $\mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right) \times \mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right) \to \mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right)$ is bilinear and (jointly) continuous.

Proof. We have

$$
\partial^\alpha (fg) = \sum_{\substack{\beta \in \mathbb{N}^d \\ \beta \leq \alpha}} \frac{\alpha!}{\beta!(\alpha-\beta)!} \partial^{\alpha-\beta} f\, \partial^\beta g
$$

so

$$
\begin{aligned}
\langle x \rangle^k \left| \partial^\alpha (fg)(x) \right| &\leq \sum_{\substack{\beta \in \mathbb{N}^d \\ \beta \leq \alpha}} \frac{\alpha!}{\beta!(\alpha-\beta)!} \|f\|_{\alpha-\beta,k} \|g\|_{\beta,0} \\
&\leq \tau_1(f) \tau_2(g)
\end{aligned}
$$

where

$$
\begin{aligned}
\tau_1(f) &= \left(\sum_{\substack{\beta \in \mathbb{N}^d \\ \beta \leq \alpha}} \max_{\beta \leq \alpha} \frac{\alpha!}{\beta!(\alpha-\beta)!} \right) \max_{\beta \leq \alpha} \|f\|_{\beta,k} \\
\tau_2(g) &= \max_{\beta \leq \alpha} \|g\|_{\beta,0}.
\end{aligned}
$$

Therefore, by the seminorm criterion for continuity of multilinear maps, it follows that $(f,g) \mapsto fg$ is continuous. □

Definition: If $f, g \in \mathcal{S}\left(\mathbb{R}^d, \mathbb{K}\right)$, then

$$
(f * g)(x) = \int_{\mathbb{R}^d} f(y) g(x-y)\, d^d y
$$

is called the convolution of two functions.

Note that

$$
\int_{\mathbb{R}^d} \left| f(y) g(x-y) \right| d^d y \leq \int_{\mathbb{R}^d} \langle y \rangle^{-(d+1)} \|f\|_{0,d+1} \|g\|_{0,0}\, d^d y,
$$

so this is well-defined.

Proposition: The convolution $f * g$ is C^∞, and for every α, $\partial^\alpha (f * g) = f * (\partial^\alpha g)$.

Proof. By induction on $|\alpha|$, we start by observing that $\left| \partial_i (f(y) g(x-y)) \right| = \left| f(y) \partial_i (g(x-y)) \right|$, where ∂_i denotes a single partial derivative with respect to the coordinate x_i. Since the derivative of any Schwartz function is Schwartz, and the product of any two Schwartz functions is Schwartz, it follows that the latter function is Schwartz, hence in L^1. Therefore, by Dominated Convergence, it follows that

$$
\partial_i \int f(y) g(x-y)\, d^d y = \int f(y) \partial_i (g(x-y))\, d^d y.
$$

By Clairaut's Theorem, we may write any n-fold partial derivative ∂_x^α as $\partial_i \partial_x^\beta$, where β is a d-dimensional

multi-index satisfying $\left|\beta\right| = |\alpha| - 1$. Since

$$\left|\partial_x^\alpha (f(y)g(x-y))\right| = \left|\partial_i\left(\partial_x^\beta (f(y)g(x-y))\right)\right|,$$

and once again the latter function is a Schwartz function since the derivative of a Schwartz function is Schwartz. Therefore, by Dominated Convergence, it follows that we may exchange derivative and integral, yielding

$$\begin{aligned}
\partial^\alpha (f*g)(x) &= \partial_x^\alpha \int_{\mathbb{R}^d} f(y)g(x-y)\, d^d y \\
&= \partial_i \partial_x^\beta \int_{\mathbb{R}^d} f(y)\partial_x^\beta (g(x-y))\, d^d y \\
&= \partial_i \int_{\mathbb{R}^d} f(y)\left(\partial_x^\beta g\right)(x-y)\, d^d y \\
&= \int f(y)\partial_i \partial_x^\beta (g(x-y))\, d^d y \\
&= \int f(y)\partial^\alpha (g(x-y))\, d^d y \\
&= f*(\partial^\alpha g).
\end{aligned}$$

□

Lemma: For all $u, v \in \mathbb{R}^d$, one has

Proof. We have

$$\begin{aligned}
\langle u+v\rangle^2 &= 1 + |u+v|^2 \\
&\leq 1 + |u|^2 + |v|^2 + 2|u||v| \\
&\leq 1 + 2|u|^2 + 2|v|^2 \\
&\leq 2 + 2|u|^2 + 2|v|^2 \\
&\leq 2\left(1+|u|^2\right)\left(1+|v|^2\right).
\end{aligned}$$

□

Theorem: The convolution $\mathcal{S}\left(\mathbb{R}^d, \mathbb{K}\right) \times \mathcal{S}\left(\mathbb{R}^d, \mathbb{K}\right) \to \mathcal{S}\left(\mathbb{R}^d, \mathbb{K}\right)$ is well-defined, bilinear, and continuous.

Proof. We have that $\partial^\alpha (f*g)$ is C^∞. Taking brackets, we get

$$\begin{aligned}
\langle x\rangle^k \left|\partial^\alpha (f*g)\right| &= \langle x\rangle^k \left|\int_{\mathbb{R}^d} f(y)(\partial_x^\alpha g)(x-y)\, d^d y\right| \\
&\leq \langle x\rangle^k \int \langle y\rangle^\ell \|f\|_{0,\ell} \langle x-y\rangle^m \|g\|_{\alpha,m}\, d^d y,
\end{aligned}$$

where ℓ and m are to be determined. Using the fact that $\langle x\rangle \leq \sqrt{2}\langle y\rangle\langle x-y\rangle$, we get

$$\langle x\rangle^k \int \langle y\rangle^\ell \|f\|_{0,\ell} \langle x-y\rangle^m \|g\|_{\alpha,m}\, d^d y \leq 2^{k/2}\|f\|_{0,\ell}\|g\|_{\alpha,m} \int_{\mathbb{R}^d} \langle y\rangle^{k-\ell} \langle x-y\rangle^{k-m}\, d^d y.$$

Therefore, if we take $m = k$ and $\ell = d + k + 1$, we obtain that the right hand side is a constant, so that the convolution is in Schwartz space. Furthermore, we have shown that $\|f*g\|_{\alpha,k} \leq K\|f\|_{0,\ell}\|g\|_{\alpha,m}$ for some constant $K \geq 0$, so that the map is bilinear and continuous. □

Proposition: Convolution on $\mathcal{S}\left(\mathbb{R}^d, \mathbb{K}\right)$ satisfies

(i) $f*g = g*f$;

(ii) $\partial(f*g) = f*(\partial^\alpha g)$;

(iii) $\partial^\alpha(f*g) = (\partial^\alpha f)*g$.

Proof.

(i) Taking a substitution of $z = x-y$ has Jacobian determinant equal to 1, so that

$$\begin{aligned} f*g(x) &= \int_{\mathbb{R}^d} f(y)g(x-y)\,d^d y \\ &= \int_{\mathbb{R}^d} f(x-z)g(z)\,d^d z \\ &= g*f. \end{aligned}$$

(ii) We showed this earlier.

(iii) By commutativity, we have

$$\begin{aligned} \partial^\alpha(f*g) &= \partial^\alpha(g*f) \\ &= g*(\partial^\alpha f) \\ &= (\partial^\alpha f)*g. \end{aligned}$$

□

Theorem: For all $f, g \in \mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right)$, one has

$$\mathcal{F}[f*g] = \mathcal{F}[f]\mathcal{F}[g].$$

Proof. Using Fubini's Theorem, we have

$$\mathcal{F}[f*g](\xi) = \int_{\mathbb{R}^d} e^{-i\langle x,\xi\rangle} \int_{\mathbb{R}^d} f(y)g(x-y)\,d^d y\,d^d x.$$

The proper estimate is found by using Tonelli's theorem so that

$$\begin{aligned} \int_{\mathbb{R}^d}\int_{\mathbb{R}^d} \left|f(y)g(x-y)\right| d^d y\,d^d x &= \int_{\mathbb{R}^d} \left|f(y)\right| \int_{\mathbb{R}^d} \left|g(x-y)\right| d^d x\,d^d y \\ &= \left(\int_{\mathbb{R}^d} \left|f(y)\right| d^d y\right)\left(\int_{\mathbb{R}^d} g(z)\,d^d z\right) \\ &< \infty. \end{aligned}$$

Therefore, Fubini's Theorem and the substitution $z = x-y$ gives

$$\begin{aligned} \mathcal{F}[f*g](\xi) &= \int_{\mathbb{R}^d}\int_{\mathbb{R}^d} e^{-i\langle \xi,y\rangle} e^{-i\langle \xi,x-y\rangle} f(y)g(x-y)\,d^d x\,d^d y \\ &= \left(\int_{\mathbb{R}^d} e^{-i\langle \xi,y\rangle} f\,d^d y\right)\left(\int_{\mathbb{R}^d} e^{-i\langle \xi,z\rangle} g(z)\,d^d z\right) \\ &= \mathcal{F}[f]\mathcal{F}[g]. \end{aligned}$$

□

Operations and Fourier Transform for $\mathcal{S}\left(\mathbb{R}^d, \mathbb{K}\right)^*$

First, we define the derivative of a distribution $\varphi \in \mathcal{S}\left(\mathbb{R}^d, \mathbb{K}\right)^*$, we want to define $\partial^\alpha \varphi$.

First, we work like physicists. We will write

$$\begin{aligned}\varphi(f) &= \langle f, \varphi\rangle \\ &= \int_{\mathbb{R}^d} \varphi(x) f(x)\, d^d x,\end{aligned}$$

where we consider $\varphi(x)$ as a generalized function associated to φ. We would then consider the "derivative" of the distribution φ by using integration by parts, so that

$$\begin{aligned}\int_{\mathbb{R}^d} \partial^\alpha \varphi(x) f(x)\, d^d x &= (-1)^{|\alpha|} \int_{\mathbb{R}^d} \varphi(x) \partial^\alpha f(x)\, d^d x \\ &= (-1)^{|\alpha|} \langle \partial^\alpha f, \varphi\rangle \\ &=: \langle f, \partial^\alpha(\varphi)\rangle.\end{aligned}$$

From now on, we will write $\mathcal{S} = \mathcal{S}\left(\mathbb{R}^d, \mathbb{K}\right)$ (with $\mathbb{K}$ most likely equal to $\mathbb{C}$) and $\mathcal{S}^* = \mathcal{S}\left(\mathbb{R}^d\right)^*$.

Definition: Given a multi-index $\alpha \in \mathbb{N}^d$, the map $\partial^\alpha \colon \mathcal{S}^* \to \mathcal{S}^*$ takes

$$\varphi \mapsto \left(f \mapsto (-1)^{|\alpha|} \langle \partial^\alpha f, \varphi\rangle\right).$$

Proposition: The map $\partial^\alpha \colon S^* \to S^*$ is continuous with respect to the strong dual topology.

Proof. We have

$$\partial^\alpha_{\mathcal{S}^*} = (-1)^{|\alpha|} \left(\partial^\alpha_{\mathcal{S}}\right)^*$$

by definition. The transpose/adjoint is strong–strong continuous. □

Now, we note that $\mathcal{S} \subseteq \mathcal{S}^*$ by the standard evaluation. If $\iota \colon \mathcal{S} \to \mathcal{S}^*$ is the corresponding injection, then we need $\partial^\alpha_{\mathcal{S}^*} \circ \iota = \iota \circ \partial^\alpha_{\mathcal{S}}$.

Proof. We recall that for any $f \in \mathcal{S}$, the definition of the duality between $\mathcal{S}$ and $\mathcal{S}^*$ is given by

$$\langle g, \iota(f)\rangle = \int g(x) f(x)\, d^d x.$$

We note that integration by parts gives

$$\begin{aligned}(-1)^{|\alpha|} \left\langle \partial^\alpha_{\mathcal{S}}(g), \iota(f)\right\rangle &= (-1)^{|\alpha|} \int_{\mathbb{R}^d} f(x) \partial^\alpha_{\mathcal{S}}(g(x))\, d^d x \\ &= \int \partial^\alpha_{\mathcal{S}}(f(x)) g(x)\, d^d x \\ &= \left\langle g, \iota\left(\partial^\alpha_{\mathcal{S}}(f)\right)\right\rangle\end{aligned}$$

□

Example: Let $d = 1$, and let

$$H(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0. \end{cases}$$

Define a distribution $\varphi \in \mathcal{S}^*$ by taking

$$\langle f, \varphi\rangle = \int_{\mathbb{R}} H(x) f(x)\, dx.$$

Notice that

$$\left|\varphi(f)\right| \leq \|f\|_{0,2} \int \langle x\rangle^{-2}\, dx,$$

so that φ is indeed continuous. We compute the derivative to be

$$\begin{aligned}\langle f, \partial\varphi\rangle &= -\langle f, \varphi\rangle \\ &= -\int_{-\infty}^{\infty} H(x) f'(x)\,dx \\ &= -\int_{0}^{\infty} f'(x)\,dx \\ &= f(0) - f(\infty) \\ &= f(0) \\ &= \langle f, \delta_0\rangle \\ &=: \underbrace{\int_{-\infty}^{\infty} \delta(x) f(x)\,dx}_{\text{physics notation}}.\end{aligned}$$

That is, we may write $\frac{d}{dx}(H(x)) = \delta(x)$ if we were physicists.

Example: We want now to find the second derivative. That is,

$$\begin{aligned}\langle f, \partial\delta\rangle &= -\langle f', \delta\rangle \\ &= -f'(0) \\ &= \underbrace{\int_{-\infty}^{\infty} \frac{d}{dx}(\delta(x)) f'(x)\,dx}_{\text{physics notation}}\end{aligned}$$

Next, we concern ourselves with multiplication of distributions by elements of $\mathcal{O}_M$.

Definition: If $\varphi \in \mathcal{S}^*$ and $g \in \mathcal{O}_M$, then we define the pointwise product $g\varphi$ by setting it to be the unique distribution such that

$$\langle g\varphi, f\rangle = \langle \varphi, gf\rangle.$$

for all $f \in \mathcal{S}$.

Note that this function is well-defined since $gf \in \mathcal{S}$ whenever $f \in \mathcal{S}$ and $g \in \mathcal{O}_M$. Notice that this is a transpose of the continuous map $f \mapsto gf \mapsto \langle \varphi, gf\rangle$, so that the map taking $\varphi \mapsto g\varphi$ is linear and continuous.

Remark: The set $\iota(\mathcal{S}) \subseteq \mathcal{S}^*$ is sequentially dense, so in particular, the derivative $\partial^\alpha_{\mathcal{S}^*}$ and pointwise product $\varphi \mapsto g\varphi$ are the unique extensions to $\mathcal{S}^*$ from their corresponding operations on $\mathcal{S}$.

Finally, we concern ourselves with Fourier transforms of elements of $\mathcal{S}^*$.

Heuristically,[I] we write

$$\langle f, \varphi\rangle = \int f(x)\varphi(x)\,dx.$$

Imagine we may construct some $\hat{\varphi}(\xi)$ as

$$\hat{\varphi}(\xi) = \int_{\mathbb{R}} e^{-i\langle x,\xi\rangle}\varphi(x)\,d^d x,$$

and "define" the distribution $\hat{\varphi}$ by

$$\langle f, \hat{\varphi}\rangle = \int \hat{\varphi}(\xi) f(\xi)\,d^d\xi.$$

[I] That is, completely and comically unrigorously.

Plugging in directly would then yield

$$\begin{aligned}\int_{\mathbb{R}^d} \hat{\varphi}(\xi) f(\xi)\, d^d\xi &= \int \int e^{-i\langle x,\xi\rangle} \varphi(x) f(\xi)\, d^d x\, d^d\xi \\ &= \int \varphi(x) \int e^{-i\langle x,\xi\rangle} f(\xi)\, d^d\xi\, d^d x \\ &= \left\langle \hat{f}, \varphi \right\rangle \\ &=: \langle f, \mathcal{F}[\varphi]\rangle.\end{aligned}$$

Definition: If $\varphi \in \mathcal{S}^*$, we define the Fourier transform of φ by

$$\langle f, \mathcal{F}[\varphi]\rangle = \langle \mathcal{F}[f], \varphi\rangle.$$

This map extends $\mathcal{F}$ on $\mathcal{S} \subseteq \mathcal{S}^*$. Notice that on $\mathcal{S}^*$, $\mathcal{F}$ is the adjoint of $\mathcal{F}$ on $\mathcal{S}$, so that $\mathcal{F}$ is continuous. Furthermore, exercising the same process on $\mathcal{F}^{-1}$ yields that $\mathcal{F}$ is a strong topological isomorphism of $\mathcal{S}^*$ onto itself.

Remark: The operator $(2\pi)^{-d/2}\mathcal{F}_{\mathcal{S}^*}|_{L^2(\mathbb{R}^d)}$, where we view $L^2\left(\mathbb{R}^d\right) \subseteq \mathcal{S}^*$, is the Fourier transform on L^2, and it is a unitary.

Example: Physicists often write

$$\int_{-\infty}^{\infty} e^{-ix\xi}\, dx = 2\pi\delta(\xi).$$

This makes no sense. We may actually interpret this as $\mathcal{F}[\varphi]$ where φ is the distribution given by the constant function 1. That is,

$$\langle f, \varphi\rangle = \int_{-\infty}^{\infty} f(x)\, dx.$$

If we take the Fourier transform of φ under this notion, we expect it to equal $2\pi\delta(\xi)$. Computing is then given by

$$\begin{aligned}\langle f, \mathcal{F}[\varphi]\rangle &= \langle \varphi, \mathcal{F}[f]\rangle \\ &= \int_{-\infty}^{\infty} \hat{f}(x)\, dx\end{aligned}$$

Multiplying by 1 twice over yields

$$\begin{aligned}&= 2\pi\left(\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{i(0)x}\hat{f}(x)\, dx\right) \\ &= 2\pi f(0) \\ &= \langle 2\pi\delta, f\rangle.\end{aligned}$$

Example: Given some $0 < \alpha < d$, physicists may write

$$\int_{\mathbb{R}^d} e^{-i\langle x,\xi\rangle} \frac{1}{|x|^\alpha}\, d^d x = \frac{\Gamma\left(\frac{d-\alpha}{2}\right)}{\Gamma\left(\frac{\alpha}{2}\right)} 2^{d-\alpha}\pi^{d/2} \frac{1}{|\xi|^{d-\alpha}}.$$

Once again, this doesn't make much sense (i.e., it's not a well-defined Lebesgue integral). Similar to the previous case, we will define an honest distribution $\varphi \in \mathcal{S}^*$ and then establish the Fourier transform of this distribution.

Let

$$\langle f, \varphi\rangle = \int_{\mathbb{R}^d \setminus \{0\}} \frac{1}{|x|^\alpha} f(x)\, dx.$$

We claim that this is a convergent Lebesgue integral. This emerges from the fact that

$$\begin{aligned}\int_{\mathbb{R}^d\setminus\{0\}} \frac{1}{|x|^\alpha}|f(x)|\,d^dx &= \int_{0<|x|\le 1} \frac{1}{|x|^\alpha}|f(x)|\,d^dx + \int_{|x|\ge 1} \frac{1}{|x|^\alpha}|f(x)|\,d^dx \\ &\le \|f\|_{0,0}\frac{2\pi^{d/2}}{\Gamma(d/2)}\int_0^1 r^{d-1-\alpha}\,dr + \|f\|_{0,d+1}\int_{|x|\ge 1}\langle x\rangle^{-(d+1)}\,d^dx \\ &< \infty.\end{aligned}$$

This inequality automatically gives that φ is an honest distribution.

Therefore, our goal is to compute the Fourier transform of φ. We start by observing that, by taking $u = t|\xi|^2$, we have that

$$\begin{aligned}\int_0^\infty e^{-t|\xi|^2}t^{\alpha/2-1}\,dt &= \int_0^\infty e^{-u}\left(\frac{1}{|\xi|^2}\right)^{\alpha/2-1}u^{\alpha/2-1}\,du \\ &= \frac{1}{|\xi|^\alpha}\int_0^\infty e^{-u}u^{\alpha/2-1}\,du \\ &= \frac{1}{|\xi|^\alpha}\Gamma\left(\frac{\alpha}{2}\right).\end{aligned}$$

Furthermore, we compute that

$$\begin{aligned}\mathcal{F}\left[e^{-a|x|^2}\right](\xi) &= \int_{\mathbb{R}^d} e^{-i\langle\xi,x\rangle - a|x|^2}\,d^dx \\ &= (2a)^{-d/2}\int e^{-i\frac{\langle u,\xi\rangle}{\sqrt{2a}}-|u|^2/2}\,d^du \\ &= (2a)^{-d/2}(2\pi)^{d/2}e^{-\frac{1}{2}\left(\frac{|\xi|^2}{2a}\right)} \\ &= \left(\frac{\pi}{a}\right)^{d/2}e^{-\frac{|\xi|^2}{4a}}.\end{aligned}$$

By the definition of $\mathcal{F}[\varphi]$, we have

$$\begin{aligned}\langle f,\mathcal{F}[\varphi]\rangle &= \langle\mathcal{F}[f],\varphi\rangle \\ &= \int_{\mathbb{R}^d\setminus\{0\}}\frac{1}{|\xi|^\alpha}\hat{f}(\xi)\,d^d\xi \\ &= \int_{\mathbb{R}^d\setminus\{0\}}\left(\frac{1}{\Gamma\left(\frac{\alpha}{2}\right)}\int_{\mathbb{R}^d}e^{-t|\xi|^2}t^{\alpha/2-1}\,dt\right)\hat{f}(\xi)\,d^d\xi \\ &= \frac{1}{\Gamma(\alpha/2)}\int_{\mathbb{R}^d\setminus\{0\}}\int_{\mathbb{R}^d}\hat{f}(\xi)e^{-t|\xi|^2}t^{\alpha/2-1}\,dt\,d^d\xi\end{aligned}$$

Observe that

$$\int_{\mathbb{R}^d}\int_0^\infty\left|\frac{1}{\Gamma(\alpha/2)}e^{-t\xi^2}t^{\alpha/2-1}\hat{f}(\xi)\right|dt\,d^d\xi \le \int_{\mathbb{R}^d\setminus\{0\}}\frac{1}{|\xi|^\alpha}\left|\hat{f}(\xi)\right|d^d\xi,$$

so the integral is finite by Tonelli's theorem. By Fubini's theorem, we may switch the order of the integrand, giving

$$= \frac{1}{\Gamma(\alpha/2)}\int_0^\infty\int_{\mathbb{R}^d\setminus\{0\}}\hat{f}(\xi)e^{-t|\xi|^2}t^{\alpha/2-1}\,d^d\xi\,dt$$

Introducing a set of measure zero gives

$$= \int_0^\infty\left(\int_{\mathbb{R}^d}e^{-t|\xi|^2}\hat{f}(\xi)\,d^d\xi\right)\frac{t^{\alpha/2-1}}{\Gamma(\alpha/2)}\,dt$$

$$= \int_0^\infty \left(\int_{\mathbb{R}^d} \overline{e^{-t|\xi|^2}} \hat{f}(\xi)\, d^d\xi \right) \frac{t^{\alpha/2-1}}{\Gamma(\alpha/2)}\, dt$$
$$= \int_0^\infty \left(\int_{\mathbb{R}^d} \overline{\mathcal{F}\left[(4t\pi)^{-d/2} e^{-\frac{|x|^2}{4t}} \right](\xi)} \hat{f}(\xi)\, d^d\xi \right) \frac{t^{\alpha/2-1}}{\Gamma(\alpha/2)}\, dt$$

By Plancherel's Theorem, we get

$$= \int \left((2\pi)^d \int_{\mathbb{R}^d} (4t\pi)^{-d/2} e^{-\frac{|x|^2}{4t}} f(x)\, d^d x \right) \frac{t^{\alpha/2-1}}{\Gamma(\alpha/2)}\, dt$$
$$= \frac{\pi^{d/2}}{\Gamma(\alpha/2)} \int_0^\infty \int_{\mathbb{R}^d} t^{\alpha/2-1-d/2} e^{-\frac{|x|^2}{4t}} f(x)\, d^d x\, dt$$

By Fubini's Theorem and deleting a measure zero set, we get

$$= \frac{\pi^{d/2}}{\Gamma(\alpha/2)} \int_{\mathbb{R}^d \setminus \{0\}} \left(\int_0^\infty e^{-\frac{|x|^2}{4t}} t^{\frac{\alpha}{2}-\frac{d}{2}-1}\, dt \right) f(x)\, d^d x.$$

Using $v = |x|^2/4t$, we get

$$= \frac{\pi^{d/2}}{\Gamma(\alpha/2)} \int_{\mathbb{R}^d \setminus \{0\}} \left(|x|^{\alpha-d} 2^{d-\alpha} \int_0^\infty v^{\frac{d-\alpha}{2}-1} e^{-v}\, dv \right) f(x)\, d^d x$$
$$= \int_{\mathbb{R}^d \setminus \{0\}} \left(\frac{\pi^{d/2} 2^{d-\alpha} \Gamma\left(\frac{d-\alpha}{2}\right)}{\Gamma(\alpha/2)} \frac{1}{|x|^{d-\alpha}} \right) f(x)\, dx$$
$$= \langle f, \mathcal{F}[\varphi] \rangle.$$

The most important example is $d = 3$ and $\alpha = 2$. Then, $\frac{1}{|\xi|^2}$ maps to $\frac{2\pi^{d/2}\Gamma(1/2)}{\Gamma(1)} \frac{1}{|x|} = \frac{2\pi^2}{|x|}$.

In particular, concerning the Poisson equation $-\Delta\varphi = \rho$, one finds that $\varphi = \mathcal{F}^{-1}\left(\frac{1}{|\xi|^2} \right) * \rho$. Therefore, in three dimensions, we have

$$\varphi(y) = \left(\frac{1}{(2\pi)^3} \frac{2\pi^2}{|x|} \right) * \rho$$
$$= \int_{\mathbb{R}^3 \setminus \{x\}} \frac{1}{4\pi |x-y|} \rho(y)\, d^3 y.$$

Normed Vector Spaces

Our main focus is going to be on Banach spaces.

First, we note that if $V_1, \ldots, V_n, W$ are locally convex topological vector spaces, then $\mathcal{L}_n(V_1, \ldots, V_n; W)$ can be equipped with the topology given by

$$\|M\| = \sup\{\tau(M(x_1, \ldots, x_n)) : \tau \text{ a continuous seminorm on } W, (x_1, \ldots, x_n) \in A_1 \times \cdots \times A_n\}.$$

In particular, for $n = 1$ and $W = \mathbb{C}$, this gives the strong topology on the dual V^*. When $V_1, \ldots, V_n, W$ are normed spaces, then the topology on $\mathcal{L}_n(V_1, \ldots, V_n; W)$ is the operator norm topology.

Proposition: If V and W are normed spaces, and $\mathcal{L}(V, W)$ is equipped with the operator norm topology, then if W is complete, so is $\mathcal{L}(V, W)$ is complete.

The Classics of Functional Analysis

We start by quickly reviewing the Baire Category Theorem and some other results about complete metric spaces.

Theorem: Let (X,d) be a complete metric space. Let $F_0 \supseteq F_1 \supseteq \cdots$ be a nested sequence of nonempty closed subsets with $\text{diam}(F_n) \to 0$. Then, there is a unique x such that

$$\{x\} = \bigcap_{n\geq 0} F_n.$$

Proof. Pick elements $x_n \in F_n$. Then, $(x_n)_n$ is Cauchy, so it converges to some point $x \in X$. □

Theorem (Baire Category Theorem)**:** Let (X,d) be a complete metric space. Suppose $(U_n)_{n\geq 0}$ is a sequence of dense open subsets of X. Then, $\bigcap_{n\geq 0} U_n$ is dense.

Proof. Let $y \in X$, and let $W \in \mathcal{O}(y)$. We must show that $W \cap \bigcap_{n\geq 0} U_n \neq \emptyset$.

Recursively, we construct two sequences $(x_n)_n \subseteq X$ and $(r_n)_n \subseteq (0,\infty)$ such that

- $r_n \leq 2^{-n}$ for all $n \geq 0$;
- $B(x_0,r_0) \subseteq U_0 \cap W$;
- $B(x_n,r_n) \subseteq U_n \cap U(x_{n-1},r_{n-1})$ for all $n \geq 1$.

Suppose for any $N \geq 0$, we have constructed $x_0,\ldots,x_N$ and $r_0,\ldots,r_N$ such that $B(x_0,r_0) \subseteq W$, for all $0 \leq n \leq N$ we have $r_n \leq 2^{-n}$, and for every $1 \leq n \leq N$ we have $B(x_n,r_n) \subseteq U_n \cap B(x_{n-1},r_{n-1})$. Then, we can construct x_{N+1} and r_{N+1} such that the conditions hold. Such an existence follows from the fact that U_{N+1} is dense.

Finally, we apply the Cantor intersection theorem to the family of closed unit balls $\{B(x_n,r_n) : n \geq 0\}$ to find $x \in \bigcap_{n\geq 0} U_n$. □

Corollary: Suppose (X,d) is a nonempty complete metric space. Suppose $(F_n)_n$ is a sequence of closed subsets. If $\bigcup_n F_n = X$, then there exists some n such that $F_n^\circ \neq \emptyset$.

Proof. Let $U_n = F_n^c$. Suppose toward contradiction that for all $n \in \mathbb{N}$, $F_n^\circ = \emptyset$. Then, $(F_n^\circ)^c = \overline{F_n^c} = X$ for each n, or that U_n are a dense family of open sets. Thus, by Baire Category Theorem, $\overline{\bigcap_n U_n} = X$. Yet, we have

$$\begin{aligned}\overline{\bigcap_n U_n} &= \overline{\left(\bigcup_{n\geq 0} F_n\right)^c} \\ &= \overline{X^c} \\ &= \emptyset,\end{aligned}$$

or that X is empty, which is a contradiction. □

To state and prove the open mapping theorem, we need some prep work.

Throughout, $(V,\|\cdot\|)$ is a normed space.

Definition: A subset $A \subseteq V$ is called symmetric if for all $x \in A$, $-x \in A$. Equivalently, $A = -A$.

Definition: A subset $A \subseteq V$ is called convex if for all $x,y \in A$ and all $t \in [0,1]$, we have that $(1-t)x + ty \in A$.

Proposition: For all $v \in V$ and all $r > 0$, the open ball $U(v,r)$ and the closed ball $B(v,r)$ are symmetric and convex.

Proposition:

(i) If $A \subseteq V$ is symmetric and convex, and $v \in V$ with $r > 0$, then $U(v,r) \subseteq A$ implies $U(0,r) \subseteq A$. We call this the "tube argument."

(ii) If $B(v,r) \subseteq A$, then $B(0,r) \subseteq A$.

Proposition: Let $A \subseteq V$ be a subset.

(i) If A is symmetric, then $\overline{A}$ is symmetric.

(ii) If A is convex, then $\overline{A}$ is convex.

Proposition: Let $\mathcal{F}$.

(i) If $A \in \mathcal{F}$ is symmetric for all A, then $\bigcap_{A\in\mathcal{F}} A$ is symmetric and $\bigcup_{A\in\mathcal{F}} A$ is symmetric.

(ii) If $A \in \mathcal{F}$ is convex for all A, then $\bigcap_{A\in\mathcal{F}} A$ is convex.

Proposition: Let V and W be two normed spaces. Let $T\colon V \to W$ be a $\mathbb{K}$-linear map.

(i) If $A \subseteq V$ is symmetric, then $T(A) \subseteq W$ is symmetric.

(ii) If $A \subseteq V$ is convex, then $T(A) \subseteq W$ is convex.

(iii) If $B \subseteq W$ is symmetric, then $T^{-1}(B) \subseteq V$ is symmetric.

(iv) If $B \subseteq W$ is convex, then $T^{-1}(B) \subseteq V$ is convex.

Definition: A map $\rho\colon V \to \mathbb{R}$ is called a convex function if for all $0 \le t \le 1$ and all $x, y \in V$, we have

$$\rho((1-t)x + ty) \le (1-t)\rho(x) + t\rho(y).$$

Proposition: The map $\rho\colon V \to \mathbb{R}$ is convex if and only if $\{(x, v) \in V \times \mathbb{R} : \lambda \ge \rho(x)\}$ is a convex subset of $V \times \mathbb{R}$.

Theorem: If V and W are Banach spaces, and $T\colon V \to W$ is a surjective continuous linear map, then T is an open map.

Proof. For each $n \ge 0$, let $F_n = \overline{T(U(0, n+1))}$. This is a closed subset of W. Since $0 \in W$, W is nonempty. Since $W = \bigcup_n F_n$, it follows by the corollary to Baire Category Theorem, there is some n such that $F_n^\circ \neq \emptyset$.

By linearity, since $F_n = (n+1)F_0$, it follows that $F_0^\circ \neq \emptyset$. Thus, there exists $w \in W$ and $r_0 > 0$ such that $U(w, r_0) \subseteq \overline{T(U(0,1))}$. Notice that $\overline{T(U(0,1))}$ is convex and symmetric. In particular, $U(0, r_0) \subseteq \overline{T(U(0,1))}$, so $B(0, r_0) \subseteq \overline{T(U(0,1))}$.

We claim that for all $w \in W$, if $\|w\| \le r_0$, then there exists $v \in V$ such that $\|v\| \le 1$ and $\|w - Tv\| \le \frac{1}{2} r_0$. In particular, we observe that any such w is an element of $\overline{T(U(0,1))}$. Thus, $U(w, r_0/2) \cap T(U(0,1)) \neq \emptyset$. Therefore, we choose such a $\widetilde{w}$ in this intersection, and let $v \in U(0,1)$ be such that $Tv = \widetilde{w}$. Furthermore, for all $n \ge 0$ and all $w \in W$ with $\|w\| \le 2^{-n} r_0$, there exists v with $\|v\| \le 2^{-n}$ and $\|w - Tv\| \le 2^{-(n+1)} r_0$. This emerges from rescaling the details from the previous paragraph.

Next, we claim that $B(0, r_0) \subseteq T(B(0,2))$. If $w_0 \in W$ is such that $\|w_0\| \le r_0$, then we see that there is v_0 with $\|v_0\| \le 1$ and $\|w_0 - Tv_0\| \le \frac{1}{2} r_0$. Recursively, we construct $(w_n)_n$ by letting $w_n = w_{n-1} - \sum_{j=0}^{n-1} Tv_j$, where we have that $\|v_j\| \le 2^{-j}$ and $\|w_k - Tv_k\| \le 2^{-(k+1)} r_0$. Notice that $w_n \to 0$ as $n \to \infty$, so that

$$w_0 = \lim_{n\to\infty} T\left(\sum_{j=0}^{n-1} v_j\right).$$

Moreover, $\sum_{j=0}^{\infty} \|v_j\| < \infty$, so $v = \sum_{j=0}^{\infty} v_j$ is defined in V since V is complete. Since T is continuous, it follows that $w_0 = Tv$. Finally, we have that $\|v\| \le \sum_{j=0}^{\infty} \|v_j\| \le 2$.

Let $U \subseteq V$ be open, $w \in T(U)$. There is $v \in U$ such that $Tv = w$. Let $\varepsilon > 0$ to be determined; we will show that $U(w, \varepsilon) \subseteq T(U)$. Notice that

$$\begin{aligned} U(w, \varepsilon) &= Tv + U(0, \varepsilon) \\ &\subseteq Tv + B(0, ve) \\ &= Tv + \frac{\varepsilon}{r_0} B(0, r_0) \end{aligned}$$

$$\begin{aligned}
&\subseteq Tv + \frac{\varepsilon}{r_0} T(B(0,2)) \\
&= T\left(B\left(v, \frac{2\varepsilon}{r_0}\right)\right) \\
&\subseteq T\left(U\left(v, \frac{3\varepsilon}{r_0}\right)\right).
\end{aligned}$$

We choose ε small enough such that $U\left(v, \frac{3\varepsilon}{r_0}\right) \subseteq U$, and obtain the desired result. □

Corollary: If $T\colon V \to W$ is a continuous linear bijection of Banach spaces, T is an isomorphism of topological vector spaces.

The next classic of functional analysis is the *closed graph theorem*.

If V and W are Banach spaces, we let $V \times W$ be equipped with the norm given by

$$\|(v,w)\| = \max(\|v\|, \|w\|).$$

Then, $V \times W$ with this norm is a Banach space.

Let $T\colon V \to W$ be a $\mathbb{K}$-linear map. The *graph* of T is

$$\text{graph}(T) = \{(v, Tv) \in V \times W : v \in V\}.$$

In particular, $\text{graph}(T)$ is a linear subspace.

Theorem: The map T is continuous if and only if $\text{graph}(T) \subseteq V \times W$ is closed in the product topology.

Proof. If T is continuous, then if $(v_n)_n \to v \in V$, we have $(Tv_n)_n \to Tv$ by definition, so $(v_n, Tv_n) \to (v, Tv) \in \text{graph}(T)$.

If $\Gamma = \text{graph}(T)$ is closed, then we define maps $\pi_1\colon V \times W \to V$ and $\pi_2\colon V \times W \to W$ are given by $\pi_1(v,w) = v$ and $\pi_2(v,w) = w$. These are continuous linear maps. Since Γ admits a norm given by the restriction from the product, and Γ is closed, it follows that Γ is a Banach space.

Notice that $\pi_1|_\Gamma\colon \Gamma \to V$ is a continuous bijection. Therefore, there is a continuous inverse $(\pi_1|_\Gamma)^{-1}$ Since

$$T = \pi_2 \circ (\pi_1|_\Gamma)^{-1},$$

it follows that T is continuous. □

Remark: Often, to prove continuity, we want to show that $(x_n)_n \to x$ implies $(Tx_n)_n \to Tx$. Using the closed graph theorem gives a free extra hypothesis that $(Tx_n)_n \to y$, so it is only necessary to show that $y = Tx$.

The next classic of functional analysis is the Uniform Boundedness Principle.

Theorem: Suppose $\{T_i : i \in I\}$ is a family of continuous linear operators $V \to W$, where V is a Banach space and W is a normed space. If it is the case that for all $x \in V$, we have $\sup_{i\in I}\|T_i x\| < \infty$ (i.e., pointwise bounded), then $\sup_{i\in I}\|T_i\| < \infty$.

Proof. For any $n \geq 0$, let

$$\begin{aligned}
F_n &= \left\{ v \in V : \sup_{i\in I} \|T_i v\| \leq n+1 \right\} \\
&= \bigcap_{i\in I} T_i^{-1}(B(0, n+1)).
\end{aligned}$$

Then, F_n is closed for each $n \geq 0$, with $\bigcup_{n\geq 0} F_n = V$. By Baire Category Theorem, there is some $n \geq 0$ such that $F_n^\circ \neq \emptyset$. Since $F_n = (n+1)F_0$, it follows that $F_0^\circ \neq \emptyset$. There exists $v \in V$ and $\varepsilon > 0$ such that $U(v,\varepsilon) \subseteq F_0$. Since F_0 is symmetric and convex, the tube argument gives that $U(0,\varepsilon) \subseteq F_0$. In particular, this means that

for any $\lambda > 0$, we have

$$U(0, \lambda\varepsilon) \subseteq \left\{ v \in V : \sup_i \|T_i v\| \leq \lambda \right\}.$$

If $\lambda = 2/\varepsilon$, then $B(0,1) \subseteq U(0,2)$, so that we get our desired result. □

Example (Hellinger–Toeplitz Theorem)**:** Let H be a Hilbert space with inner product $\langle \cdot, \cdot \rangle$. The convention for this class is that we will let the inner product be conjugate linear in the first coordinate.

Let $A : H \to H$ be a symmetric operator. Then, A is continuous.

One way we can prove this is by using the closed graph theorem. Suppose $(x_i)_i$ is a net in H converging to some $x \in H$.

Hahn–Banach Theorems

In order to discuss extending linear functionals, we start by assuming that the vector space V is separable, so we won't need the axiom of choice.

We start by discussing a result in linear algebra. Suppose V is a $\mathbb{K}$-vector space, and $S \subseteq V$. Then, we recall that

$$\begin{aligned} \operatorname{span}(S) &= \bigcap \{ W : S \subseteq W, W \text{ is a linear subspace} \} \\ &= \left\{ \sum_{\text{finite}} \lambda_i w_i : \lambda_i \in \mathbb{C}, w_i \in S \right\}. \end{aligned}$$

The first definition is what we might call an "external definition," while the second is what we might call an "internal definition."

Definition: A partially ordered set $(X, \preceq)$ is called

- totally ordered if for all $x, y \in X$, $x \preceq y$ or $y \preceq x$;
- well-ordered if and only if for all $\emptyset \neq S \subseteq X$, $\min(S)$ exists;
- a subset $I \subseteq X$, where $(X, \preceq)$ is well-ordered, is called an initial segment if for all $x \in I$ and $y \in X$, $y \preceq x$ implies that $y \in I$.

Example: The natural numbers $\mathbb{N}$ are well-ordered under the standard additive ordering, and an initial segment is of the form $\{0, 1, \ldots, N-1\}$ or is equal to all of $\mathbb{N}$.

Lemma (Algorithmic Cleanup Lemma)**:** Let V be a vector space, W a subspace, and $(u_n)_{n \in \mathbb{N}}$ a sequence in V. Then, there exists an initial segment $I \subseteq \mathbb{N}$ and $(v_n)_{n \in \mathbb{N}}$ such that

(i) $\operatorname{span}(W \cup \{u_n : n \in \mathbb{N}\}) = \operatorname{span}\{W \cup \{v_n : n \in I\}\}$;

(ii) for all $n \in I$, $v_n \notin \operatorname{span}(W \cup \{v_m : m < n\})$.

Proof. For any $n \in \mathbb{N}$, let $X_n = \operatorname{span}(W \cup \{u_m : m < n\})$. Let

$$J = \{ n \in \mathbb{N} : u_n \notin X_n \}.$$

Let $Y_n = \operatorname{span}(W \cup \{u_m : m \in J, m < n\})$. We claim that for all $n \in \mathbb{N}$, $X_n = Y_n$.

We proceed by "ill-founded induction" (i.e., no base case). Let $n \in \mathbb{N}$, and suppose for all $m \in \mathbb{N}$ with $m < n$, we have $X_m = Y_m$. We will show that $X_n = Y_n$. Let $w \in X_n$. Then, w is a finite linear combination of some elements of W and elements u_m with $m \in J$ and $m < n$. Observe that $W \subseteq Y_n$. For such an m, then either

- $m \in J$ and $m < n$, meaning $u_m \in Y_n$, so $w \in Y_n$;
- $m \notin J$ and $m < n$, so by the way we defined J, $u_m \in X_m$, which equals Y_m by the hypothesis; since $Y_m \subseteq Y_n$, we once again have $w \in Y_n$.

Thus, the claim is proven.

Now, if $J = \{n \in \mathbb{N} : u_n \notin X_n\}$, then there is I an initial segment of $\mathbb{N}$ and a strictly increasing bijection $\varphi\colon I \to J$. Then, we may define $v_n = u_{\varphi(n)}$ to get the desired result. □

Remark: To generalize strong induction to properties P_i defined over well-ordered sets I, we let

$$B = \{i \in I : P_i \text{ is false}\},$$

and if B is nonempty, then there is a least element i_0. That is, for all $i < i_0$, P_i holds. If we show the general statement "if P_i holds for any $i < j$, then P_j holds," then we obtain a contradiction because then it would follow that i_0.

Proposition (One-Step Extension)**:** Let V be a $\mathbb{R}$-vector space, $W \leq V$ a subspace, $v \in V \setminus W$, $L\colon W \to \mathbb{R}$ an $\mathbb{R}$-linear functional, and $p\colon V \to \mathbb{R}$ a convex function. Suppose $L \leq p$ on W. Then, there is $\widetilde{L}\colon W \oplus \mathbb{R}v \to \mathbb{R}$ an $\mathbb{R}$-linear form such that $\widetilde{L}|_W = L$ and $\widetilde{L} \leq p$ on $W \oplus \mathbb{R}v$.

Proof. Let $\alpha \in \mathbb{R}$. Define $\widetilde{L}\colon W \oplus \mathbb{R}v \to \mathbb{R}$ given by $w + \lambda v \mapsto L(w) + \lambda\alpha$. This is an $\mathbb{R}$-linear well-defined map that is an extension of L.

Our goal is to show now that $\widetilde{L} \leq p$. In particular, this is equivalent to showing that the set of all viable α is non-empty; that is,

$$L(w) + \lambda\alpha \leq p(w + \lambda v).$$

Splitting into three statements, we have the following.

- If $\lambda = 0$, we have that $L(w) \leq p(w)$, which is true by hypothesis.
- If $\lambda > 0$, then

$$\alpha \leq \frac{1}{\lambda}(p(w + \lambda v) - L(w)).$$

- If $\lambda < 0$, then we let $\mu = -\lambda$, so we have

$$\alpha \geq \frac{1}{\mu}(L(w) - p(w - \mu v)).$$

We will be done if we have $S \leq I$, where

$$S = \sup_{\substack{\mu>0 \\ w\in W}} \frac{1}{\mu}(L(w) - p(w - \mu v))$$
$$I = \sup_{\substack{\lambda>0 \\ w\in W}} \frac{1}{\lambda}(p(w + \lambda v) - L(w)).$$

Let $\mu, \lambda > 0$ and $w_1, w_2 \in W$. Then, we have

$$\begin{aligned}
\lambda L(w_1) + \mu L(w_2) &= (\lambda + \mu)L\left(\frac{\lambda}{\lambda+\mu}w_1 + \frac{\mu}{\lambda+\mu}w_2\right) \\
&\leq (\lambda + \mu)p\left(\frac{\lambda}{\lambda+\mu}w_1 + \frac{\mu}{\lambda+\mu}w_2\right) \\
&= (\lambda + \mu)p\left(\frac{\lambda}{\lambda+\mu}(w_1 - \mu v) + \frac{\mu}{\lambda+\mu}(w_2 + \lambda v)\right) \\
&\leq (\lambda + \mu)\left(\frac{\lambda}{\lambda+\mu}p(w_1 - \mu v) + \frac{\mu}{\lambda+\mu}p(w_2 + \lambda v)\right)
\end{aligned}$$

Dividing all by $\lambda\mu$, we have

$$\frac{1}{\mu}(L(w_1) - p(w_1 - \mu v)) \leq \frac{1}{\lambda}(p(w_2 + \lambda v) - L(w_1)).$$

This yields the desired result. □

Remark: Folland and others work by taking p to be a sublinear functional — that is, for every $x, y \in V$, $p(x+y) \leq p(x) + p(y)$ and for every $t \geq 0$, $p(tx) = tp(x)$. This automatically implies convexity.

Theorem (Hahn–Banach Extension for Separable Real Normed Spaces)**:** Let V be a separable real normed space. Let $W \leq V$ be a linear subspace, p be a continuous convex function on V, and $L\colon W \to \mathbb{R}$ such that $L \leq p$ on W. Then, there exists $\widetilde{L}\colon V \to \mathbb{R}$ such that $\widetilde{L}|_W = L$ and $\widetilde{L} \leq p$ on V.

Proof. Let $(u_n)_n$ be a countable dense subset of V. Then, $W_\infty = \operatorname{span}(W \cup \{u_n : n \in \mathbb{N}\})$ is a dense linear subspace of V. Use the clean-up lemma such that I is an initial segment of $\mathbb{N}$ and a collection $\{v_n : n \in I\}$ in V such that

$$W_\infty = \operatorname{span}(W \cup \{v_n : n \in N\}).$$

Furthermore, for all $n \in I$, we have that $v_n \notin \operatorname{span}(W \cup \{v_m : 1 \leq m < n\})$. Write $W_{<n}$ for the latter set, and let $W_n = \operatorname{span}(W_{<n} \cup \{v_n\})$. Then, we observe that if $n_1, n_2 \in I$ with $n_1 \leq n_2$, we have $W_{n_1} \leq W_{n_2}$ and $W_{<n_1} \leq W_{<n_2}$. Furthermore,

$$W_\infty = \bigcup_{n \in I} W_n.$$

We will construct $L_\infty\colon W_\infty \to \mathbb{R}$ which is $\mathbb{R}$-linear, extends L, and $L_\infty \leq p$ on W_∞.

This will be done by constructing $(L_n)_{n\in I}$ such that $L_n\colon W_n \to \mathbb{R}$ is $\mathbb{R}$-linear, extends L, and $L_n \leq p$ on W_n. Recursively, we let $n \in I$ with L_m constructed such that for all $m \leq n$ in I, the following compatibility condition holds: for all $k \in I$ with $k \leq m$, $L_m|_{W_k} = L_k$. Notice that $W_{<n} = \bigcup\{W_m : m \leq n \text{ with } n \in I\}$. Let $L_{<n}\colon W_{<n} \to \mathbb{R}$ be defined by $x \mapsto L_m(x)$ if $x \in W_m$ for some $m \in I$ with $m < n$.

If $W_n = W_{<n} \oplus \mathbb{R}v_n$, then the one-step extension yields that there is $L_n\colon W_n \to \mathbb{R}$ that is $\mathbb{R}$-linear with $L_n|_{W_{<n}} = L_{<n}$ with $L_n \leq p$ on W_n, and the compatibility condition is satisfied for any $m < n$ along with $m = n$.

This yields our desired sequence $(L_n)_{n\in I}$ such that $L_n\colon W_n \to \mathbb{R}$ satisfies $L_n|_W = L$, $L_n \leq p$, and for every $m_1, m_2 \in I$ with $m_1 \leq m_2$, we have $L_{m_2}|_{W_{m_1}} = L_{m_1}$.

Define $L_\infty\colon \bigcup_{n\in I} W_n \to \mathbb{R}$ be defined by $L_\infty(x) = L_n(x)$ if $x \in W_n$ for some $n \in I$.

We claim that there exists $K \geq 0$ such that for all $x \in W_\infty$, $|L_\infty(x)| \leq K\|x\|$. Observe that since p is a continuous convex function, there is some constant $C > 0$ such that for any $\varepsilon > 0$, $\|x\| \leq \varepsilon$ implies that $p(x) \leq C$.

If $x = 0$, then $L_\infty(x) = 0$, while if $x \neq 0$ with $\|x\| \leq \varepsilon$, then

$$\begin{aligned} L_\infty(x) &= \frac{\|x\|}{\varepsilon} L_\infty\left(\frac{\varepsilon}{\|x\|}x\right) \\ &\leq \frac{\|x\|}{\varepsilon} p\left(\frac{\varepsilon}{\|x\|}x\right) \\ &\leq \frac{\|x\|}{\varepsilon} C. \end{aligned}$$

We apply to $-x$ also, so that $|L_\infty(x)| \leq K\|x\|$, where $K = \frac{C}{\varepsilon}$. Thus, L_∞ is continuous, so that L_∞ extends to $\overline{W_\infty} = V$ with the same norm. □

Corollary: Let $(V, \|\cdot\|)$ be a separable real normed vector space, and let $W \leq V$ a linear subspace. Suppose $L\colon W \to \mathbb{R}$ is a continuous $\mathbb{R}$-linear form with respect to the topology on V. Then, there is a continuous

extension $\widetilde{L}\colon V \to \mathbb{R}$ such that $\widetilde{L}|_W = L$ and $\left\|\widetilde{L}\right\| = \|L\|$.

Proof. We will define $p\colon V \to \mathbb{R}$ by $p(x) = \|L\|\|x\|$. Then, p is a continuous convex function on V. Then, for every $x \in W$, we have $|L(x)| \leq \|x\|\|L\|$.

By Hahn–Banach extension, there is $\widetilde{L}\colon V \to \mathbb{R}$ such that $\widetilde{L}|_W = L$ and $\widetilde{L} \leq p$ on V. That is, for every $x \in V$, we have $\widetilde{L}(x) \leq \|L\|\|x\|$. An analogous result holds for $-x$, so that $\left|\widetilde{L}(x)\right| \leq \|L\|\|x\|$. Thus, $\left\|\widetilde{L}\right\| \leq \|L\|$, while we automatically have

$$\begin{aligned}\|L\| &= \sup_{x\in W} \frac{\|L(x)\|}{\|x\|} \\ &= \sup_{x\in W} \frac{\left\|\widetilde{L}(x)\right\|}{\|x\|} \\ &\leq \sup_{x\in V} \frac{\left\|\widetilde{L}(x)\right\|}{\|x\|} \\ &= \left\|\widetilde{L}\right\|.\end{aligned}$$

□

The Riesz Representation Theorem

Our focus will be on the version of the Riesz Representation Theorem for compact metrizable spaces.

Let $(X, \mathcal{T}_X)$ be a topological space. We will write $\mathcal{B}_X$ for the Borel σ-algebra on X — i.e., $\sigma(\mathcal{T}_X)$. We will assume that X is metrizable. Our goal is to study positive Borel measures on X; that is, $\mu\colon \mathcal{B}_X \to [0, \infty]$.

Definition: Let (X, d) be a metric space. Let $A \subseteq X$. Then, we define the diameter

$$\operatorname{diam}(A) \coloneqq \sup_{x,y\in A} d(x, y).$$

In particular, if $A \subseteq B$, then $\operatorname{diam}(A) \subseteq \operatorname{diam}(B)$.

If $x \in X$ and $A \subseteq X$, then we define the distance from x to A by

$$\operatorname{dist}(x, A) = \inf_{y\in A} d(x, y).$$

Notice that if $x, y \in X$ and $z \in A$, we have that

$$\begin{aligned}\operatorname{dist}(x, A) &\leq d(x, z) \\ &\leq d(x, y) + d(y, z),\end{aligned}$$

so we obtain that for all $x, y \in A$, $\left|\operatorname{dist}(x, A) - \operatorname{dist}(y, A)\right| \leq d(x, y)$, meaning that the distance function is continuous.

Proposition: If A is closed, then $d(x, A) = 0$ if and only if $x \in A$.

Proposition (Urysohn's Lemma for Metric Spaces)**:** Let A and B be nonempty, closed, and disjoint. Then, there exists $f : X \to \mathbb{R}$ continuous such that $f(x) \in [0, 1]$, $f|_A = 1$, and $f|_B = 0$.

Proof. Take

$$f = \frac{\operatorname{dist}(x, B)}{\operatorname{dist}(x, A) + \operatorname{dist}(x, B)}.$$

□

Remark: In fact, $f(x) = 1$ if and only if $x \in A$, and similarly, $f(x) = 0$ if and only if $x \in B$.

The notion of bumpfunctionology is a smooth version of the Urysohn lemma.

From now on, assume that X is a nonempty compact metrizable space with compatible metric d. Further-

more, we will assume that $\mathbb{K} = \mathbb{R}$, and so $C(X)$ is the space of real-valued continuous functions on X. Let $\mu\colon \mathcal{B}_X \to [0, \mu(X)]$ be a finite Borel measure on X with $\mu(X) < \infty$.

Define $\mathcal{R}[\mu]\colon C(X) \to \mathbb{R}$ given by

$$\mathcal{R}[\mu](f) = \int_X f \, d\mu.$$

This map is well-defined, $\mathbb{R}$-linear, and continuous. The continuity emerges from the fact that for any $f \in C(X)$, we have

$$\begin{aligned}\left|\int f \, d\mu\right| &\leq \int |f| \, d\mu \\ &\leq \|f\|\mu(X),\end{aligned}$$

so that $\mathcal{R}[\mu] \in C(X)^*$ is continuous of operator norm at most $\mu(X)$. Furthermore, we have

$$\mathcal{R}[\mu](1) = \mu(X),$$

so $\left\|\mathcal{R}[\mu]\right\|_{\text{op}} = \mu(X)$.

Definition: Let $L\colon C(X) \to \mathbb{R}$ be an $\mathbb{R}$-linear form. We say L is positive if for all $f \geq 0$, one has $L(f) \geq 0$.

Proposition: Let $L\colon C(X) \to \mathbb{R}$ be a positive. Then, L is positive if and only if L is continuous and $\|L\|_{\text{op}} = L(1)$.

Proof. Suppose L is positive. Then, $\|f\|1 \leq f \leq \|f\|1$, so that $\|f\|L(1) \leq L(f) \leq \|f\|L(1)$. Therefore, $|L(f)| \leq \|f\|L(1)$. Thus, L is continuous with norm at most $L(1)$; taking $f = 1$ yields the desired outcome.

Let $f \geq 0$. By the hypothesis, for all $g \in C(X)$ we have $|L(g)| \leq \|g\|L(1)$. Note that $0 \leq f \leq \|f\|1$. In particular, this means that $0 \leq \|f\|1 - f \leq \|f\|1$. Take $g = \|f\|1 - f$. Then, we have that $\|g\| \leq \|f\|$. Therefore, we have that $L(g) \leq \|g\|L(1) \leq \|f\|L(1)$. Yet, substituting the definition of f, we have

$$\begin{aligned}L(\|f\|1 - f) &= \|f\|L(1) - L(f) \\ &\leq \|f\|L(1),\end{aligned}$$

so that $L(f) \geq 0$. □

We have a map

$$\mathcal{R}\colon \{\text{finite regular positive Borel measures on } X\} \to \{\text{positive } \mathbb{R}\text{-linear forms on } C(X)\}$$

taking $\mu \mapsto \mathcal{R}[\mu]$.

Theorem (Riesz Representation Theorem)**:** The map $\mathcal{R}$ is a bijection.

We will first show that $\mathcal{R}$ is an injection. We will let (X, d) be a compact metric space. Let μ be a finite Borel measure on X. For any $A \subseteq X$ (not necessarily Borel), define

$$\mu^*(A) = \inf\{\mu(U) : U \text{ open}, A \subseteq U\}.$$

Note that $0 \leq \mu^*(A) \leq \mu(X)$.

Proposition: The map $\mu^*\colon \mathcal{P}(X) \to \mathbb{R}$ is an outer measure.

Proof. Observe that $\emptyset$ is open, so $\mu^*(\emptyset) = 0$.

If $A \subseteq B$, then for any $U \supseteq B$, we have $U \subseteq A$, so that $\mu^*(A) \subseteq \mu^*(B)$.

Finally, suppose $\{A_n : n \in \mathbb{N}\}$ is a sequence of subsets. Let $\varepsilon > 0$. For any $n \in \mathbb{N}$, there is $U_n \supseteq A_n$ such that

$\mu(U_n) \leq \mu^*(A_n) + \varepsilon 2^{-(n+1)}$. Let $W = \bigcup_n U_n$. Then,

$$\begin{aligned}\mu^*\left(\bigcup_{n=0}^{\infty} A_n\right) &\leq \mu(W) \\ &\leq \sum_{n=0}^{\infty} \mu(U_n) \\ &\leq \sum_{n=0}^{\infty} \left(\mu^*(A_n) + \varepsilon 2^{-(n+1)}\right) \\ &\leq \sum_{n=0}^{\infty} \mu^*(A_n) + \varepsilon.\end{aligned}$$

Since $\varepsilon > 0$, we obtain our desired result. □

Recall that if $A \subseteq X$ is any subset, then A is μ^*-measurable if and only if for any $E \subseteq P(X)$, we have $\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$. That is, A is a good "cookie cutter" for μ^*.

Proposition: We have A is μ^*-measurable if and only if $\mu^*(X) = \mu^*(A) + \mu^*(A^c)$.

Proof. We only need to show that $\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c)$ assuming this property.

Let $U, V, W \subseteq X$ be open such that $A \subseteq U$, $A^c \subseteq V$, and $E \subseteq W$, and let $\varepsilon > 0$. Our goal is to show that given such a W, one can pick U and V such that $\mu(W) \geq \mu(W \cap U) + \mu(W \cap V) - \varepsilon$.

Notice that if C and D are Borel, then $\mu(C)+\mu(D) = \mu(C \cup D)+\mu(C \cap D)$. Taking $C = W \cap U$ and $D = W \cap V$, then

$$\begin{aligned}\mu(W \cap U) + \mu(W \cap V) &= \mu(W \cap (U \cup V)) + \mu(W \cap U \cap V) \\ &= \mu(W) + \mu(W \cap U \cap V),\end{aligned}$$

where we use the fact that $A \subseteq U$ and $A^c \subseteq V$ so that $U \cup V = X$. Our amended goal is to find W such that $\mu(W \cap U \cap V) \leq \varepsilon$. In fact, it is enough to pick U and V such that $\mu(U \cap V) \leq \varepsilon$.

To find this, we pick U such that $\mu(U) \leq \mu^*(A) + \varepsilon/2$ and V such that $\mu(V) \leq \mu^*(A^c) + \varepsilon/2$. Then,

$$\begin{aligned}\mu(U \cap V) &= \mu(U) + \mu(V) - \mu(U \cup V) \\ &= \mu(U) + \mu(V) - \mu(X).\end{aligned}$$

Therefore, we have

$$\begin{aligned}\mu(U \cap V) &\leq \mu^*(A) + \varepsilon/2 + \mu^*(A^c) + \varepsilon/2 - \mu(X) \\ &= (\mu^*(A) + \mu^*(A^c) - \mu(X)) + \varepsilon \\ &= \varepsilon.\end{aligned}$$

This yields our desired result. □

Definition: If $A \subseteq X$ is an arbitrary subset, then we define

$$\mu_*(A) = \sup\{\mu(K) : K \text{ compact}, K \subseteq A\}.$$

Proposition: For every $A \subseteq X$, $\mu_*(A) = \mu(X) - \mu^*(A^c)$.

So, A is μ^*-measurable if and only if $\mu^*(A) = \mu_*(A)$. In particular, for every $A \subseteq \mathcal{B}_X$, we have $\mu_*(A) \leq \mu(A) \leq \mu^*(A)$. In particular, by Carathéodory's Theorem, we have that the μ^*-measurable subsets form a σ-algebra.

Proposition: If $U \subseteq X$ is open, then $\mu_*(U) = \mu(U) = \mu^*(U)$.

Proof. We have the inequalities $\mu_*(U) \leq \mu(U) = \mu^*(U)$ since U is Borel and U is a satisfactory for $\mu^*(U)$.

Our goal is to show that $\mu_*(U) \geq \mu(U)$.

If $U = X$, then we are done. If $U \neq X$, then U^c is closed and non-empty, so we have access to the function $x \mapsto \text{dist}(x, U^c)$. For any $n \geq 1$, let $K_n = \{x \in X : \text{dist}(x, U^c) \geq 1/n\}$.

The sets K_n are closed (hence compact), nested, and satisfy

$$\begin{aligned}\bigcup_{n\geq 1} K_n &= \{x \in X : \text{dist}(x, U^c) > 0\} \\ &= \{x \in X : \text{dist}(x, U^c) = 0\}^c \\ &= (U^c)^c \\ &= U.\end{aligned}$$

Therefore, we have

$$\begin{aligned}\mu_*(U) &\geq \sup_n \mu(K_n) \\ &= \mu\left(\bigcup_{n\geq 1} K_n\right) \\ &= \mu(U)\end{aligned}$$

by continuity from below. □

Therefore, the σ-algebra of μ^*-measurable subsets of X contains all the open subsets of X, so it contains $\mathcal{B}_X$. In particular, we have shown the following theorem.

Theorem: Let $(X, \mathcal{T}_X)$ be a compact metrizable space. Let μ be a finite Borel measure on X. Then, for every $A \in \mathcal{B}_X$, we have

$$\begin{aligned}\mu(A) &= \inf\{\mu(U) : U \supseteq A \text{ open}\} &(*)\\ &= \sup\{\mu(K) : A \supseteq K \text{ compact}\}. &(**)\end{aligned}$$

We say that a measure that satisfies $(*)$ is *outer regular* and a measure that satisfies $(**)$ is *inner regular*. A measure that is both outer regular and inner regular is called regular (or Radon).[II]

We note that we will only be focused on compact metrizable spaces. There are versions of the Riesz Representation Theorem that apply to locally compact Hausdorff spaces that may not be metrizable, such as $[0,1]^{\mathbb{R}}$ and $\mathbb{R}^{\mathbb{R}}$.

Remark: During the development of measure theory, there was a schism of the approach towards the subject. Those that focused on locally compact Hausdorff spaces primarily studied on functional analysis/operator algebras (results such as Gelfand duality are applicable in the setting of locally compact Hausdorff spaces), while those that focused on Polish spaces primarily studied descriptive set theory and probability.

We will show now that the map $\mu \mapsto \mathcal{R}[\mu]$ is injective.

Proposition: Let (X, d) be a compact metric space, μ a finite Borel measure, and let $\mathcal{R}[\mu] = L\colon C(X) \to \mathbb{K}$ be defined by

$$L(f) = \int f \, d\mu.$$

Then, for all $U \subseteq X$ open,

$$\mu(U) = \sup\{L(f) : 0 \leq f \leq \chi_U\}.$$

[II] On a locally compact Hausdorff space, one requires that a Radon measure has $\mu(K) < \infty$ for all compact subsets of X.

Proof. First, we observe that, by definition,

$$\begin{aligned}\int f\,d\mu &\leq \int \chi_U\,d\mu \\ &= \mu(U),\end{aligned}$$

so that the supremum is less than or equal to $\mu(U)$.

In the reverse direction, we split into three cases. If $\mu = \emptyset$, then we are done. If $\mu = X$, then $f = 1$ is a continuous function, so we are done.

If $\emptyset \neq U \subsetneq X$, then $U^c \neq \emptyset$. So, we let K_n be the aforementioned exhaustion of U by compact subsets. When n is large enough, $K_n \neq \emptyset$ since U^c is nonempty. By Urysohn's Lemma, there is a function $f_n \colon X \to [0,1]$ continuous such that $f_n|_{K_n} = 1$ and $f_n|_{U^c} = 0$. Therefore, for large enough n, we have

$$\begin{aligned}\mu(K_n) &\leq L(f_n) \\ &\leq \mu(U).\end{aligned}$$

By continuity from below, since $\sup_n \mu(K_n) = \mu(U)$, it follows that

$$\mu(U) = \sup\{L(f) : 0 \leq f \leq \chi_U\}.$$

□

Thus, the map $\mu \mapsto \mathcal{R}[\mu]$ is injective.

In order to show surjectivity, given some linear functional L, we consider the "definition" of $\mu \colon \mathcal{B}_X \to [0,\infty]$ given by

$$\begin{aligned}\mu(U) &= \sup\{L(f) : 0 \leq f \leq \chi_U\} \\ \mu(A) &= \inf\{\mu(U) : U \supseteq A \text{ open}\}\end{aligned}$$

for any Borel subset $A \subseteq X$. This is the way it is proven in classic references such as Folland's *Real Analysis* and Rudin's *Real and Complex Analysis*.

However, we will prove it a different way.

We start by showing the Riesz representation theorem for the Cantor set, $\Delta = \{0,1\}^{\mathbb{N}}$. The Cantor surjection theorem gives a surjective continuous map $\Delta \to X$. The final step is to transfer the theorem from Δ to X by using Hahn–Banach.

Theorem: The map $\mathcal{R}$ is surjective when $X = \{0,1\}^{\mathbb{N}}$.

Proof. Give X the metric given by

$$d(x,y) = \sum_{n=0}^{\infty} 2^{-n}\left|x_n - y_n\right|.$$

This defines the topology on X. Moreover, X is complete under this metric. We will show that X is compact by showing that X is totally bounded. Let $\varepsilon > 0$, and let $N \geq 0$. If $I = (i_0, \ldots, i_N) \in \{0,1\}^{N+1}$, we let

$$x_I = (i_0, \ldots, i_N, 0, 0, \ldots)$$

in X. Suppose $y = (y_n)_n$, let $I = (y_0, \ldots, y_N)$, and we observe that

$$\begin{aligned}d(x_I, y) &= \sum_{n=N+1}^{\infty} 2^{-n}\left|y_n\right| \\ &\leq 2^{-N}.\end{aligned}$$

Therefore, we may choose N large enough such that $2^{-N} < \varepsilon$, so that the balls of radius ε centered at the x_I cover X.

We say a subset $A \subseteq X$ is a *cylinder set* if one may write

$$A = \prod_{i=0}^{\infty} A_i$$

with $A_i \subseteq \{0,1\}$ and all but finitely many A_i are not equal to $\{0,1\}$. That is, A is the finite intersection of inverse images of coordinate projections.

Notice that if A is a cylinder set, then A is open (by the definition of the product topology) and closed (since every subset in the discrete topology is closed and the projection maps are continuous), so each A is compact. Let

$$\mathcal{C} = \{A \subseteq X : A \text{ is a cylinder set}\}.$$

Define

$$\mathcal{A} = \left\{ \bigsqcup_{j=1}^{n} A_j : A_j \in \mathcal{C} \right\}.$$

We will show that $\mathcal{A}$ is an algebra of subsets. First, we start by showing that $\mathcal{C}$ is an *elementary family*. That is, $\emptyset \in \mathcal{C}$, $A \cap B \in \mathcal{C}$ if $A, B \in \mathcal{C}$, and for any $A \in \mathcal{C}$, A^c is the finite disjoint union of elements of $\mathcal{C}$.

- Observe that $\emptyset = \pi_0^{-1}(\emptyset)$.
- A finite intersection of finite intersections of inverse images of coordinate projections is a finite intersection of inverse images of coordinate projections.
- Let

  $$A = A_0 \times A_1 \times \cdots \times A_N \times \prod_{j=N+1}^{\infty} \{0,1\}.$$

  Observe that

  $$A^c = \bigsqcup_{i=0}^{N} A_0 \times \cdots \times A_{i-1} \times A_i^c \times \{0,1\} \times \cdots.$$

Since $\mathcal{A}$ is the collection of finite disjoint unions of an elementary family $\mathcal{C}$, we have that $\mathcal{A}$ is an algebra of subsets.

Let $L\colon C(X) \to \mathbb{R}$ be a positive linear functional. Then, we define $\mu(A) = L(\chi_A)$. Since A is clopen, χ_A is continuous. Now, $mu(\emptyset) = L(0) = 0$, and if $A = A_1 \cup \cdots \cup A_n$ is a finite disjoint union of elements of $\mathcal{A}$, we observe that

$$\begin{aligned} \mu(A) &= L(\chi_A) \\ &= \sum_{j=1}^{n} L\left(\chi_{A_j}\right) \\ &= \sum_{j=1}^{n} \mu\left(A_j\right). \end{aligned}$$

Finally, we show that μ is σ-additive on $\mathcal{A}$ with respect to countable disjoint families. Suppose we have $A = \bigcup_{n=0}^{\infty} A_n$ and $A \in \mathcal{A}$. Since A is a finite union of cylinder sets as $A \in \mathcal{A}$, A is compact (it is the finite union of closed subsets of the compact set X). The family of A_n is an open cover of A since each A_n is clopen, so that $A = \bigcup_{n=0}^{N} A_n$. In particular, $A_n = \emptyset$ for each $n > N$. Thus,

$$\mu(A) = \sum_{n=0}^{N} \mu(A_n)$$

$$= \sum_{n=0}^{\infty} \mu(A_n).$$

Thus, μ on $\mathcal{A}$ extends to a measure on $\sigma(\mathcal{A})$. Our goal now is to show that $\sigma(\mathcal{A})$ contains the Borel σ-algebra. However, the cylinder sets are a basis for the topology on X, so in particular, this means that $\mathcal{T}_X \subseteq \sigma(\mathcal{A})$ since $\mathcal{C}$ is countable.

Thus, we have a measure $\mu\colon \mathcal{B}_X \to [0,\infty]$ such that $\mu(X) = L(1) < \infty$. Now, let $f \in C(X)$. For any $N \geq 0$, let $f_N\colon X \to \mathbb{R}$ be given by

$$f_N = \sum_{I \in \{0,1\}^{N+1}} f(i_0,\ldots,i_N,0,\ldots)\chi_{X_I},$$

where $X_I = \{i_0\} \times \cdots \times \{i_N\} \times \{0,1\}^{\mathbb{N}}$. Notice that each X_I is a cylinder set, hence in A. This is a finite sum, so that

$$\int f_N \, d\mu = L(f_N).$$

Since $f_N \to f$ uniformly, we get our desired result. □

Proposition: The space $\left(C\left(\{0,1\}^{\mathbb{N}},\mathbb{R}\right),\|\cdot\|\right)$ is a separable normed space.

Proof. If $f\colon \{0,1\}^{\mathbb{N}} \to \mathbb{R}$ is continuous, then we know that f can be approximated by the family of f_N, where

$$f_N = \sum_{\{0,1\}^{N+1}} f(i_0,\ldots,i_N,0,0,\ldots)\chi_{\{i_0\}\times\cdots\times\{i_n\}\times\{0,1\}^{\mathbb{N}}}.$$

uniformly.

Therefore, we take all the elements of $\mathbb{Q}^{<\mathbb{N}}$, which are finitely supported functions of the form

$$\sum_{\{0,1\}^{N+1}} \lambda_{i_0,\ldots,i_N}\chi_{\{i_0\}\times\cdots\times\{i_N\}\times\{0,1\}^{\mathbb{N}}},$$

where $N \geq 0$ and λ are in $\mathbb{Q}$. This is a countable dense subset. □

Theorem (Hausdorff–Alexandroff)**:** Let X be a compact metrizable space. Then, there exists $\Phi\colon \{0,1\}^{\mathbb{N}} \to X$ such that Φ is surjective and continuous.

Proof. Pick a compatible metric d on X.

We start by taking

$$X = \bigcup_{x \in X} B(x,1),$$

where $B(x,1)$ denotes the closed ball. These are of diameter 2. The open balls $\bigcup_{x\in X} U(x,1)$ cover X, so that there are $x_1,\ldots,x_{m_0}$ such that

$$X = \bigcup_{i_0=1}^{m_0} U\left(x_{i_0},1\right).$$

Thus, if we define $B_{i_0}^{(0)} = B\left(x_{i_0},1\right)$ for each $i_0 \in [m_0]$, this defines our base case.

Let $N \geq 0$. Suppose we have found $m_0,\ldots,m_N \geq 1$ and sets

$$\mathcal{B}_0 = \left\{B_{i_0}^{(0)} : i_0 \in [m_0]\right\}$$

$$\mathcal{B}_1 = \left\{ B^{(1)}_{i_0,i_1} : (i_0,i_1) \in [m_0]\times[m_1] \right\}$$
$$\vdots$$
$$\mathcal{B}_N = \left\{ B^{(N)}_{i_0,\dots,i_N} : (i_0,\dots,i_N) \in [m_0]\times\cdots\times[m_N] \right\}$$

such that all of these sets are nonempty, compact, and for all $0 \le M \le N$, $\operatorname{diam}\left(B^{(M)}_{i_0,\dots,i_M}\right) \le 2^{-M+1}$ for all $(i_0,\dots,i_M) \in [m_0]\times\cdots\times[m_M]$. Furthermore, for all M with $0 \le M < N$ and $(i_0,\dots,i_M) \in [m_0]\times\cdots\times[m_M]$, we have that

$$B^{(M)}_{i_0,\dots,i_M} = \bigcup_{i_{M+1}=1}^{m_{M+1}} B^{(M+1)}_{i_0,\dots,i_M,i_{M+1}} \tag{$*$}$$

Fix $(i_0,\dots,i_N) \in [m_0]\times\cdots\times[m_N]$. The open balls $U\left(x, 2^{-N-1}\right)$ for each $x \in B^{(N)}_{i_0,\dots,i_N}$ cover the subset, so there are $m_{N;i_0,\dots,i_N}$ and $x_1,\dots,x_{m_{N;i_0,\dots,i_N}}$ such that

$$B^{(N)}_{i_0,\dots,i_N} \subseteq \bigcup_{i_{N+1}=1}^{m_{N;i_0,\dots,i_N}} U\left(x_{i_{N+1}}, 2^{-N-1}\right)$$

For $i_{N+1} \in \left[m_{N;i_0,\dots,i_N}\right]$, let

$$B^{(N+1)}_{i_0,\dots,i_N,i_{N+1}} = B^{(N)}_{i_0,\dots,i_N} \cap B\left(x_{i_{N+1}}, 2^{-N-1}\right). \tag{$**$}$$

This is nonempty, compact, and has diameter at most 2^{-N} the criterion ($*$). Let

$$m_{N+1} = \max\left\{ m_{N;i_0,\dots,i_N} : (i_0,\dots,i_N) \in [m_0]\times\cdots\times[m_N] \right\}$$

In the case that $i_{N+1} > m_{N;i_0,\dots,i_N}$, we define $B^{(N+1)}_{i_0,\dots,i_{N+1}} = B^{(N+1)}_{i_0,\dots,i_N,1}$, and leave the definition as in ($**$).

Let $Y = \prod_{j=0}^{\infty}\left[m_j\right]$, equipped with the discrete topology. Let $y = (i_0,i_1,\dots) \in Y$. Then, by the Cantor Intersection Theorem,

$$\Psi(y) = \bigcap_{N\ge 0} B^{(N)}_{i_0,i_1,\dots,i_N}$$

is well-defined since we are taking an intersection of nonempty nested subsets of a complete metric space with decreasing diameter.

Let $x \in X$. Then,

$$x \in \bigcup_{i=0}^{m_0} B^{(0)}_{i_0},$$

so we choose i_0 to be the smallest such that $x \in B_{i_0}$. Writing

$$B^{(0)}_{i_0} = \bigcup_{i_1=1}^{m_1} B^{(1)}_{i_0,i_1},$$

we choose the smallest i_1 such that $x \in B^{(1)}_{i_0,i_1}$, and continue along this process. Thus, we get $y = (i_0,i_1,\dots)$ such that $\Psi(y) = x$.

For continuity, if $y = (i_0, i_1, \dots)$ and $z = (j_0, j_1, \dots)$ coincide until index N, then $d(\Psi(y), \Psi(z)) \leq 2^{-N+1}$. Recall that the subset of Y consisting of all $z \in Y$ that agree with y up to some index forms a neighborhood of y, so that Ψ is continuous.

Finally, for each N, we pick d_N to be the smallest integer at least 0 such that $m_N \leq 2^{d_N}$, and pick

$$\tau_N \colon \{0,1\}^{d_N} \to [m_N]$$

to be the binary representation. Then, the identification

$$\{0,1\}^{\mathbb{N}} \cong \prod_{j=0}^{\infty} \{0,1\}^{d_j}$$

yields a continuous surjection $\{0,1\}^{\mathbb{N}} \to Y$, whence $\Phi \colon \{0,1\}^{\mathbb{N}} \to X$ is our desired map. □

The following statements are going to be on homework assignments.

Proposition: Let $f : X \to Y$.

(i) If $\mathcal{N}$ is a σ-algebra on Y, then

$$f^*\mathcal{N} = \left\{A \subseteq X : \text{ there exists } B \in \mathcal{N} \text{ such that } A = f^{-1}(B)\right\}$$

is a σ-algebra on X.

(ii) If $\mathcal{M}$ is a σ-algebra on X, then

$$f_*\mathcal{M} = \left\{B \subseteq Y : f^{-1}(B) \in \mathcal{M}\right\}$$

is a σ-algebra on Y.

Proposition: Suppose $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ are measurable spaces, and $f : X \to Y$ is a map. The following are equivalent:

(i) f is $\mathcal{M}$-$\mathcal{N}$-measurable;

(ii) $\mathcal{M} \supseteq f^*\mathcal{N}$;

(iii) $\mathcal{N} \subseteq f_*\mathcal{M}$.

Now, suppose $(X, \mathcal{M}, \mu)$ is a measure space and $(Y, \mathcal{N})$ is a measurable space, and let $f : X \to Y$ be a $\mathcal{M}$-$\mathcal{N}$-measurable map. Then,

$$f_*\mu(B) := \mu\left(f^{-1}(B)\right)$$

is a measure on $(Y, \mathcal{N})$.

Theorem (Abstract Change of Variables)**:** If $g : Y \to \mathbb{C}$ is a measurable function and $f : X \to Y$ is as above, then

$$\int_Y g \, d(f_*\mu) = \int_X g \circ f \, d\mu.$$

The proof is by bootstrapping.

Theorem: Let X be a compact metrizable space, and let L be a positive linear form on $C(X, \mathbb{R})$. Then, there is a positive Borel measure μ on X such that

$$L(f) = \int f \, d\mu.$$

Proof. Since X is compact metrizable, there is some continuous surjection $\Phi\colon \{0,1\}^{\mathbb{N}} \to X$. Let $\Gamma\colon C(X,\mathbb{R}) \to C\left(\{0,1\}^{\mathbb{N}},\mathbb{R}\right) =: V$ be given by

$$\Gamma(f) = f \circ \Phi.$$

Observe that for every $f \in C(X,\mathbb{R})$, we have that $\|\Gamma(f)\| = \|f\|$. In particular, Γ is injective (since it is an isometry).

Let $\Gamma(C(X,\mathbb{R})) = W \subseteq V$. This is a separable subspace. Let

$$\begin{aligned} L_W\colon W &\to \mathbb{R} \\ g &\mapsto L\left(\Gamma^{-1}(g)\right). \end{aligned}$$

This is a well-defined linear form on W. If $W \ni g \neq 0$, then

$$\begin{aligned} \frac{|L_W(g)|}{\|g\|} &= \frac{L\left(\Gamma^{-1}(g)\right)}{\left\|\Gamma^{-1}(g)\right\|} \\ &\leq \|L\|, \end{aligned}$$

so that $\|L_W\| = \|L\|$ So, L_W is continuous on W. Moreover, $\|L\| = L(\chi_X)$. Moreover,

$$\Gamma(\chi_X) = \chi_{\{0,1\}^{\mathbb{N}}},$$

so that $\chi_{\{0,1\}^{\mathbb{N}}} \in W$. Using the Hahn–Banach extension, we have some linear form $L_V\colon V \to \mathbb{R}$ with $L_V|_W = L_W$ and $\|L_V\| = \|L_W\|$. Since

$$\begin{aligned} L_V\left(\chi_{\{0,1\}^{\mathbb{N}}}\right) &= L_W\left(\chi_{\{0,1\}^{\mathbb{N}}}\right) \\ &= \|L_W\| \\ &= \|L_V\|, \end{aligned}$$

it follows that L_V is a positive linear form. Thus, there is some Borel measure ν on $\{0,1\}^{\mathbb{N}}$ such that for all $g \in V$,

$$L_V(g) = \int_{\{0,1\}^{\mathbb{N}}} g \, d\nu.$$

Defining $\mu = \Phi_* \nu$ gives us our desired result.

Our goal now is to show that $\mathcal{R}[\mu] = L$. Let $f \in C(X,\mathbb{R})$. Then,

$$\begin{aligned} L(f) &= L\left(\Gamma^{-1} \circ \Gamma(f)\right) \\ &= L_W(\Gamma(f)) \\ &= L_V(\Gamma(f)) \\ &= \int_{\{0,1\}^{\mathbb{N}}} \Gamma(f) \, d\nu \\ &= \int_{\{0,1\}^{\mathbb{N}}} f \circ \Phi \, d\nu \\ &= \int_X f \, d(\Phi_* \mu) \\ &= \int_X f \, d\mu \\ &= \mathcal{R}[\mu](f) \end{aligned}$$

□

Remark:

(i) We recall that if f is differentiable, then we may write

$$f(x) = f(0) + \int_0^x f'(t)\,dt.$$

Constructing measures, however, is quite difficult. Thanks to the Riesz Representation Theorem, we have an addition to the Carathéodory criterion to construct measures.

(ii) We now have a dictionary between measures and linear forms on continuous functions. In the previous proof, the Hahn–Banach theorem essentially allowed us to take a measure μ on X and define a measure ν on $\{0,1\}^{\mathbb{N}}$ such that $\Phi_*\nu = \mu$. In a sense, this allowed us to define a pullback (or lift) of a measure on X to a measure on $\{0,1\}^{\mathbb{N}}$. This is related to a variety of questions.

- The coarea formula in Riemannian geometry, which asks how we may define volume forms on e.g. S^{d-1}.
- How we may extend integration to a Lie group G from integration on H and on the homogeneous space G/H.
- Regular conditional probabilities, which ask how we may define probabilities $P(A|B)$ when $P(B) = 0$, such as trying to find $E[f|X = x]$.
- The disintegration theorem — given some measure μ on X, and some map $f : X \to Y$, whether we may define measures μ^y on Y that are supported on the fibers $f^{-1}(\{y\})$.

(iii) Recall that $(X, \mathcal{T})$ is a Polish space if X is separable and there is a compatible complete metric on X. It's possible to show that for any Polish space X, there is a continuous surjection $\Phi : \mathbb{N}^{\mathbb{N}} \to X$, where $\mathbb{N}^{\mathbb{N}}$ is called the *Baire space*. A space is called *Souslin* if there exists any continuous surjection $\Phi : \mathbb{N}^{\mathbb{N}} \to X$, and we have shown that Polish spaces are Souslin.

Bochner's Theorem

Definition: Let $\psi : \mathbb{R}^d \to \mathbb{C}$. We say ψ is a function of positive (definite) type if for every $n \geq 0$ and all $\xi_1, \ldots, \xi_n \in \mathbb{R}^d$, and for all $z_1, \ldots, z_n \in \mathbb{C}$,

$$\sum_{i,j=1}^{n} \overline{z_i} z_j \psi\Big(\xi_i - \xi_j\Big) \in [0, \infty).$$

Equivalently, for all $\xi_1, \ldots, \xi_n \in \mathbb{R}^d$, the matrix $\Big(\psi\Big(\xi_i - \xi_j\Big)\Big)_{i,j}$ is positive.

Definition: Let $\mu \in \text{Prob}\Big(\mathbb{R}^d\Big)$ be a Borel probability measure. The characteristic function $\Phi_\mu : \mathbb{R}^d \to \mathbb{C}$ is given by

$$\xi \mapsto \int_{\mathbb{R}^d} e^{i\langle \xi, x\rangle}\, d\mu(x).$$

In the homework, it was shown that Φ_μ is continuous at 0 and $\Phi_\mu(0) = 1$, and Φ_μ is of positive type. Bochner's Theorem shows the converse.

Theorem (Bochner's Theorem)**:** If $\psi : \mathbb{R}^d \to \mathbb{C}$ is a function of positive type with $\psi(0) = 1$ and ψ continuous at 0, then there is a unique probability measure μ such that

$$\psi(\xi) = \int_{\mathbb{R}^d} e^{i\langle \xi, x\rangle}\, d\mu(x).$$

Definition: We say a sequence of probability measures $(\mu_n)_n$ converges to a probability measure μ in the

weak topology if for all $f \in C^b(X, \mathbb{R})$, one has

$$\lim_{n\to\infty} \int f \, d\mu_n = \int f \, d\mu.$$

In the homework, it was shown that if two functions Φ_μ and Φ_ν coincide, then $\mu = \nu$.

Notice that if ψ is a function of positive type, then for all $\eta \in \mathbb{R}^d$, one has $\psi(-\eta) = \overline{\psi(\eta)}$ and $\left|\psi(\eta)\right| \leq \psi(0)$. That this holds emerges from the fact that the matrix $\left(\psi\left(\xi_i - \xi_j\right)\right)_{i,j}$ is Hermitian and positive, so one has $\psi\left(\xi_i - \xi_j\right) = \overline{\psi\left(\xi_j - \xi_i\right)}$ and, when considering 0 and one vector ξ, one has $\psi(0)^2 - \left|\psi(\xi)\right|^2 \geq 0$ as the matrix has positive determinant.

Finally, we also have that

$$\left|\psi(\alpha+\beta) - \psi(\alpha)\right|^2 \leq 2\psi(0)\operatorname{Re}(\psi(0) - \psi(\beta))$$

for any $\alpha, \beta \in \mathbb{R}^d$. Furthermore, this inequality implies that if ψ is continuous at 0 and of positive type, then ψ is continuous everywhere.

We note that the map $\psi := \delta_0$ is a function of positive type, but it is not continuous at 0.

First, we show the existence part of Bochner's Theorem. We will seek to define

$$\mu(A) = \int \chi_A \, d\mu,$$

or equivalently, an integral

$$\int f \, d\mu$$

for all bounded measurable functions f. Using the Riesz theorem, it is satisfactory to define such a measure for satisfactory continuous functions, and using density, it is satisfactory to do so for Schwartz functions. However, because $\mathbb{R}^d$ is not compact, we will work on the one-point compactification $\mathbb{R}^d \cup \{\infty\}$.

Proof of Existence. Let ψ be a function of positive type such that ψ is continuous at 0 an $\psi(0) = 1$.

Note that, since $\left|\psi(\xi)\right| \leq \psi(0) = 1$ for all $\xi \in \mathbb{R}^d$, we have that ψ is an element of $C^b\left(\mathbb{R}^d\right)$.

Let

$$\begin{aligned} T : \mathcal{S}\left(\mathbb{R}^d\right) &\to \mathbb{C} \\ f &\mapsto \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \hat{f}(\xi)\psi(\xi) \, d^d\xi. \end{aligned}$$

This is a well-defined linear map since $\hat{f} \in \mathcal{S}\left(\mathbb{R}^d\right)$ and ψ is continuous and bounded. Moreover,

$$|Tf| \leq \frac{1}{(2\pi d)} \int_{\mathbb{R}^d} \langle\xi\rangle^{-d+1} \|f\|_{0,d+1} \, d^d\xi.$$

Thus, T is continuous, so $T \in \mathcal{S}\left(\mathbb{R}^d\right)^*$.

We will show now that T is a positive linear functional. First, we will show that for all $f \in \mathcal{S}\left(\mathbb{R}^d\right)$,

$$\int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \overline{f}(\xi) f(\eta) \psi(\xi - \eta) \, d^d\xi \, d^d\eta \geq 0. \tag{$*$}$$

We note that

$$\langle(\xi_1, \ldots, \xi_d, \eta_1, \ldots, \eta_d)\rangle \leq \sqrt{2}\langle\xi\rangle\langle\eta\rangle,$$

so that

$$\langle(\xi_1,\dots,\xi_d,\eta_1,\dots,\eta_d)\rangle\left|\overline{f(\xi)}f(\eta)\psi(\xi-\eta)\right| \leq 2^{k/2}\|f\|_{0,k}^2.$$

Therefore, $\overline{f(\xi)}f(\eta)\psi(\xi-\eta)$ has rapid decay. In particular, the function $\mathbb{R}^{2d}\to\mathbb{C}$ taking $(\xi,\eta)\mapsto\overline{f(\xi)}f(\eta)\psi(\xi-\eta)$. The function $\overline{f(\xi)}f(\eta)\psi(\xi-\eta)$ may not be in the Schwartz space, even though it is continuous.

So,

$$G_{r,s}=L^{2dr}\sum_{\xi,\eta\in\Lambda_{r,s}^{2d}}\overline{f(\xi)}f(\eta)\psi(\xi-\eta)$$

satisfies

$$\lim_{s\to\infty}\lim_{r\to-\infty}G_{r,s}=\Theta.$$

Here, $\Lambda_{r,s}^d=L^r\mathbb{Z}^d\cap\left[-\frac{L^s}{2},\frac{L^s}{2}\right)^d$, where L is an integer at least 3. That is, we are taking a Riemann sum (in a sense) over $\mathbb{R}^d$. In particular, $\Lambda_{r,s}^{2d}\cong\Lambda_{r,s}^d\times\Lambda_{r,s}^d$, so this construction splits nicely. In particular, since ψ is of positive type, it follows that that Θ, hence the integral $(*)$ is positive.

Next, we will show that for every $f\in\mathcal{S}\left(\mathbb{R}^d\right)$, $\overline{T(f)}=T\left(\overline{f}\right)$. We have

$$\begin{aligned}
\overline{T(f)}&=\frac{1}{(2\pi)^d}\int_{\mathbb{R}^d}\overline{\hat{f}(\xi)}\overline{\psi(\xi)}\,d^d\xi\\
&=\frac{1}{(2\pi)^d}\int_{\mathbb{R}^d}\overline{\mathcal{F}[f](\xi)}\psi(-\xi)\,d^d\xi\\
&=\frac{1}{(2\pi)^d}\int_{\mathbb{R}^d}\mathcal{F}\left[\overline{f}\right](-\xi)\psi(-\xi)\,d^d\xi\\
&=\frac{1}{(2\pi)^d}\int_{\mathbb{R}^d}\mathcal{F}\left[\overline{f}\right](\xi)\psi(\xi)\,d^d\xi\\
&=T\left(\overline{f}\right).
\end{aligned}$$

Now, we evaluate

$$\int_{\mathbb{R}^d}\int_{\mathbb{R}^d}\overline{f(\xi)}g(\eta)\,d^d\xi\,d^d\eta$$

in terms of T. First, we observe that

$$\langle(\xi_1,\dots,\xi_d,\eta_1,\dots,\eta_d)\rangle^k\left|\overline{f(\xi)}g(\eta)\psi(\xi-\eta)\right|\leq 2^{k/2}\|f\|_{0,k}\|g\|_{0,k},$$

where $k=2d+1$, so the argument is integrable, and we may use Fubini's Theorem. Using change of variables of $u=\xi-\eta$, we get

$$\begin{aligned}
\int_{\mathbb{R}^d}\int_{\mathbb{R}^d}\overline{f(\xi)}g(\eta)\,d^d\xi\,d^d\eta&=\int\left(\int f(u+\eta)g(\eta)\psi(u)\,d^du\right)d^d\eta\\
&=\int\psi(u)\int\overline{f(u+\eta)}g(\eta)\,d^d\eta\,d^du.
\end{aligned}$$

Taking the change of variables of $v=-\eta$,

$$=\int\psi(u)\int\overline{f(u-v)}g(-v)\,d^dv\,d^du.$$

Writing $g(-v) = \check{g}(v)$, we have

$$= \int \psi(u)\overline{f} * \check{g}(u)\, d^d u,$$

and taking $h = \mathcal{F}^{-1}\left(\overline{f} * \check{g}\right)$

$$= (2\pi)^d T(h)$$
$$= (2\pi)^d T\left(\overline{\mathcal{F}^{-\check{1}}[f]}\mathcal{F}^{-\check{1}}[g]\right).$$

Therefore, we see that for all $f \in \mathcal{S}\left(\mathbb{R}^d\right)$, we have

$$T\left(|f|^2\right) = T\left(\overline{f} f\right)$$
$$\in [0, \infty).$$

Given $y \in \mathbb{R}^d$, let $\tau_y \colon \mathcal{S}\left(\mathbb{R}^d\right) \to \mathcal{S}\left(\mathbb{R}^d\right)$ be given by $\tau_y f(x) = f(x-y)$. Then, τ_y is a continuous linear map. This emerges from the fact that

$$\langle x\rangle^k \left|\partial^\alpha\left(\tau_y[f](x)\right)\right| = \langle x\rangle^k \left|\partial^\alpha f(x-y)\right|$$
$$\leq \|f\|_{\alpha.k} 2^{k/2}\langle y\rangle^k.$$

□